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碩士論文

有效利用能量的逐漸趨近式類比數位
轉換器設計

**Design of Energy Efficient
Successive-Approximation
Analog-to-Digital Converter**

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中華民國九十六年七月

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Design of Energy Efficient Successive-Approximation Analog-to-Digital Converter

by

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有效利用能量的逐漸趨近式類比數位轉換器設計

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摘要

本論文主要實現了兩個逐漸趨近式類比數位轉換器。在第一個轉換器中，我們利用調整開關的數量大小來得到新的電阻電容時間常數，藉以提升類比數位轉換器的速度。此外，此一設計使用自我時間控制的方法去控制比較器，這樣可以節省一半左右的功率消耗。實驗結果顯示，利用 TSMC 0.13 μm 的製程，這個 8 位元每秒取樣兩千七百萬次的逐漸趨近式類比數位轉換器平均消耗功率為 385 μW ，每次轉換所消耗的平均能量約為 105 fJ。

於第二個轉換器中，我們採用新的電容串架構，再加上被動式電荷重新分布的機制來提升轉換器的速度，同時降低電容串部分的功率消耗，而且在轉換器輸入端的電容值也可以明顯的減小。這個 8 位元每秒取樣五千萬次的被動式電荷重新分布逐漸趨近式類比數位轉換器同樣利用 TSMC 0.13 μm 的製程來進行設計，平均消耗功率為 294 μW ，每次轉換所消耗平均能量約為 41 fJ。

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Design of Energy Efficient Successive-Approximation Analog-to-Digital Converter

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ABSTRACT

In this thesis, two successive-approximation (SAR) analog-to-digital converters (ADCs) are proposed. In the first ADC, a novel RC time constant capacitor array, which adjusts the numbers of switch, is used to speed up the conversion rate. And a self timing controller is used to control the comparator, which can save half of the power consumption of the comparator. Simulation results of the first ADC, an 8-bit 27 MS/s SAR ADC, show that the total power consumption is 385 μ W and the average energy consumption per conversion step is 105 fJ in the TSMC 0.13 μ m process.

In order to further increase the conversion rate, a novel capacitor array with

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passive charge-sharing (PCS) technique, which can effectively reduce the total power consumption and the input capacitance, is used in the second ADC. Simulation results of the second ADC, an 8-bit 50 MS/s PCS SAR ADC, show that the total power consumption is 294 μW and the average energy consumption per conversion step is 41 fJ in the TSMC 0.13 μm process.

誌謝

從兩年前對類比電路的懵懂無知到今天可以完成這本論文，這路上有著很多對我伸出援手。首先我要感謝我的爸媽，在我求學的路上他們總是默默的支持的我，除了經濟上面的支持之外，他們給我很大的自由可以自己做決定，他們總是相信我可以把自己決定的事情做好，讓我可以沒有負擔的自由發展。

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Contents

Abstract

List of Figures

List of Tables

Chapter 1 Introduction

1.1 Motivation	01
1.2 Energy Efficient of Conventional ADCs	02
1.3 Thesis Organization	03

Chapter 2 The Principle and Architectures of Successive Approximation A/D Converters

2.1 Principle of Successive Approximation Algorithm	05
2.2 The Architecture of Successive Approximation ADC	07
2.2.1 DAC-Based Successive Approximation	07
2.2.2 Unipolar Charge-Redistribution A/D	07
2.2.3 Signed Charge-Redistribution A/D with a Single Reference Voltage ..	11
2.3 Capacitive Digital-to-Analog Converter	14
2.3.1 Binary-Weighted Capacitor Array	14
2.3.2 C-2C Capacitor Array	15
2.3.3 Pseudo C-2C Capacitor Array	16
2.4 Latched Comparator	18
2.4.1 Static Latched Comparator	19
2.4.2 Class-AB Latched Comparator	20
2.4.3 Dynamic Latched Comparator	21

Chapter 3 An 8-Bit 27MS/s Successive Approximation A/D Converter

3.1 Motivation	22
3.2 The Architecture of Proposed Successive Approximation A/D Converter ..	23
3.2.1 Sample and Hold	25
3.2.2 Comparator with Rail-to-Rail Range	25
3.2.3 Digital-to-Analog Converter (DAC)	28
3.3 Simulation Results	30
3.3.1 Functional simulation	30
3.3.2 Pre-Layout Simulation Result	31
3.3.3 Post-Layout Simulation Result	35

Chapter 4 Energy Efficient Passive Charge-Sharing (PCS) Successive Approximation A/D Converter

4.1 Motivation	40
4.2 The Architectures of Conventional Passive Charge-Sharing SAR ADC	41
4.2.1 Capacitor Array	43
4.2.2 Comparator	44
4.2.3 Digital Control Circuit	45
4.3 The Architectures of Proposed Passive Charge-Sharing SAR ADC	46
4.3.1 Capacitor Array	48
4.3.2 Comparator	49
4.3.3 Digital Control Circuit	51
4.4 Simulation Results	52
4.4.1 Functional Simulation	52
4.4.2 Pre-Layout Simulation	53
Chapter 5 Conclusion and Future Works	
5.1 Conclusion	56
5.2 Future Works	57
Reference	

List of Figure

Chapter 1

Figure 1-1 The ADC topologies in terms of resolution and sampling rate.....	01
Figure 1-2 Conventional architectures for Nyquist ADCs	02

Chapter 2

Figure 2-1 Flow graph for the successive-approximation approach	06
Figure 2-2 D/A converter-based successive-approximation converter.....	07
Figure 2-3 Flow graph for a modified successive approximation	08
Figure 2-4 A 5-bit unipolar charge-redistribution A/D converter.....	10
Figure 2-5 A 5-bit signed charge-redistribution A/D converter.....	13
Figure 2-6 Fully differential binary weighted capacitor array	14
Figure 2-7 Ideal C-2C capacitor array	15
Figure 2-8 Real C-2C capacitor array.....	16
Figure 2-9 Pseudo C-2C capacitor array.....	17
Figure 2-10 Thevenin equivalent of pseudo C-2C capacitor array	18
Figure 2-11 Static latched comparator	19
Figure 2-12 class-AB latched comparator	20
Figure 2-13 dynamic latched comparator	21

Chapter 3

Figure 3-1 Digital television architecture	22
Figure 3-2 Conventional unipolar charge-redistribution SAR ADC	23
Figure 3-3 Schematic of the proposed 8-bit SAR ADC	24
Figure 3-4 Schematic of the S/H circuit	25
Figure 3-5 Conventional latched comparator with one differential pairs	26
Figure 3-6 Proposed comparator with a rail-to-rail common mode input range	27
Figure 3-7 Proposed comparator with self timing controller	27
Figure 3-8 Control waveforms of self timing controller	27
Figure 3-9 Binary-weighted switched-capacitor array.....	28
Figure 3-10 Conventional RC time constant model of capacitor array	29
Figure 3-11 Modified RC array model of capacitor array.....	29
Figure 3-12 Functional simulation result.....	30
Figure 3-13 FFT simulation result of the output code with $f_{in} = 2.7\text{MHz}$ and 27MS/s conversion rate (SNDR = 49.01dB and ENOB = 7.85 bit).....	31
Figure 3-14 ENOB with respect to input frequency at 27MS/s	33
Figure 3-15 Floor plan of common centroid capacitor array	35

Figure 3-16 Layout of the capacitor array	36
Figure 3-17 Layout of the SAR and digital control circuits.....	36
Figure 3-18 Layout of the sample and hold	37
Figure 3-19 Floor-plan of the proposed ADC.....	38
Figure 3-20 Layout of the chip (core).....	38
Figure 3-21 FFT simulation result of the output code with $f_{in} = 2.7\text{MHz}$ and 27MS/s conversion rate (SNDR = 44.6dB and ENOB = 7.12 bit).....	39
Figure 3-22 Photograph of the whole ADC ($0.2 \times 0.2 \text{ mm}^2$)	39

Chapter 4

Figure 4-1 Basic charge-sharing SAR ADC architecture.....	41
Figure 4-2 Charge-sharing SAR ADC waveforms.....	42
Figure 4-3 Capacitor array and control waveform.....	44
Figure 4-4 Comparator circuit schematic	45
Figure 4-5 Proposed passive charge-sharing SAR ADC.....	47
Figure 4-6 Capacitor array of proposed passive charge-sharing SAR ADC	47
Figure 4-7 Proposed charge-sharing SAR ADC control waveforms.....	48
Figure 4-8 Proposed comparator with pre-amplifier.....	49
Figure 4-9 Work function at the comparator input node	50
Figure 4-10 Conversion of the 5 LSBs without pre-amplifier	50
Figure 4-11 Conversion of the 5 LSBs with pre-amplifier.....	51
Figure 4-12 Conversional waveform of proposed ADC	52
Figure 4-13 FFT simulation result of the output code with $f_{in} = 5\text{MHz}$ and 50MS/s conversion rate (SNDR = 45.0dB and ENOB = 7.18 bit).....	53

List of Table

Chapter 3

Table 3-1 Corner deviation simulation	32
Table 3-2 Temperature deviation simulation.....	32
Table 3-3 Supply voltage deviation simulation.....	32
Table 3-4 Power consumption of each component	33
Table 3-5 The ADC specification	34

Chapter 4

Table 4-1 Corner deviation simulation	54
Table 4-2 Power consumption of each component	54
Table 4-3 The specification of the 8-bit 50MS/s PCS SAR ADC	55

Chapter 1

Introduction

1.1 Motivation

Analog-to-digital converter provides the link between the analog world and digital system. A/D converter is often the bottleneck in a data processing application, limiting the overall speed or resolution. Figure 1-1 shows a common usage of ADC topologies in terms of resolution and sampling rate [11].

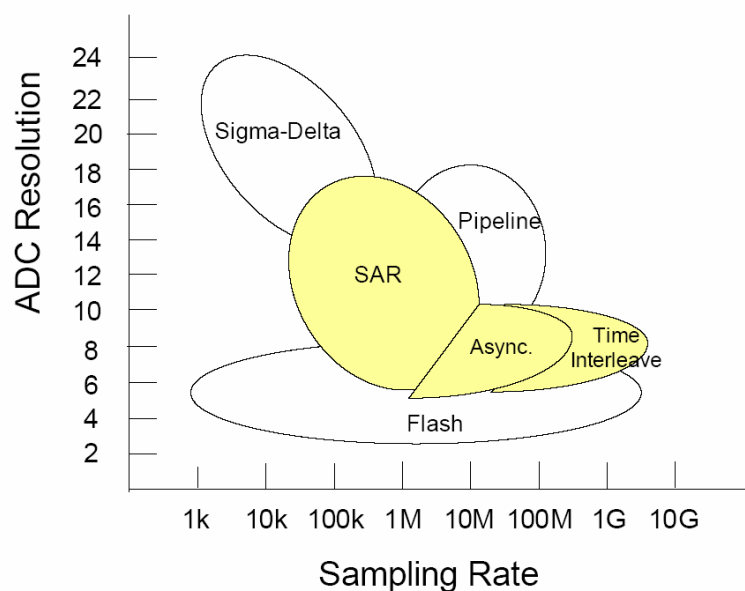


Figure 1-1 The ADC topologies in terms of resolution and sampling rate [11]

1.2 Energy Efficient of Conventional ADCs

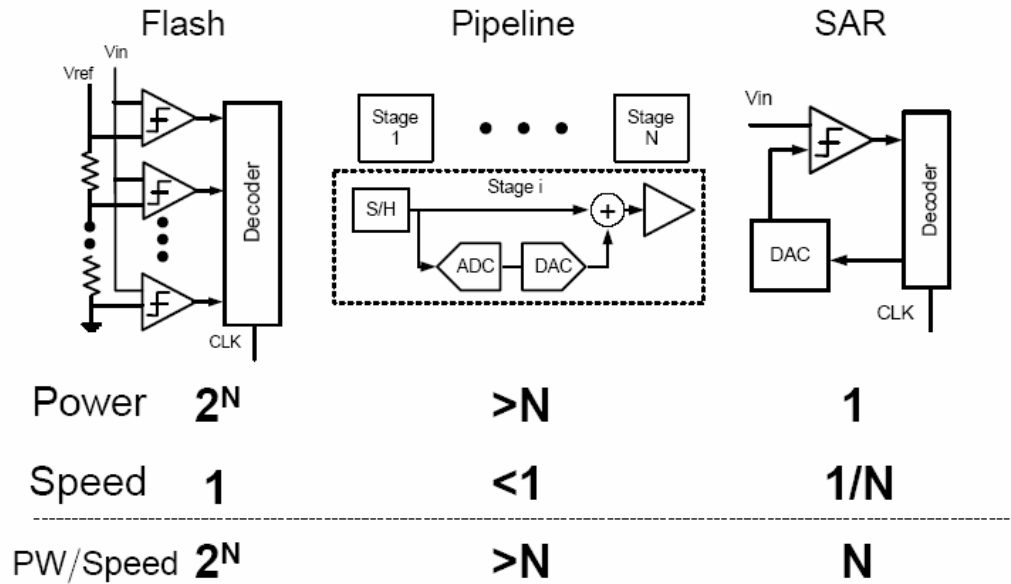


Figure 1-2 Conventional architectures for Nyquist ADCs [11]

Traditional, flash-type ADCs, including sub-ranging and folding converters, dominate high-speed and medium-resolution applications. However, flash ADCs are favored for high-speed N -bit converter since 2^N-1 comparators are utilized to make a fully parallel comparison with the entire quantization levels within one clock cycle. The decoding circuits solving sparkle and metastability issues and thermometer-to-binary code conversion also dissipate extra power. The total power consumption of a flash ADC therefore roughly scales as 2^N . In Figure 1-2, the conversion speed is normalized to one for comparison with other architectures corresponding to the fact that the full conversion is complete within one sample clock cycle. [11]

An approach to breaking the exponential dependence of the number of comparators on resolution is the use of a pipeline ADC. Instead of fully parallel

comparison, it divides the conversion into several comparison stages. Therefore, the total number of comparators is greatly reduced, only N comparators required for a 1-bit per stage, N -bit pipeline ADC. However, for the pipelined structure inter-stage residue amplification is needed which consumes considerable power and limits high speed operation. While it is possible to make use of open-loop residue amplification [17], an extra calibration loop is needed, increasing overall complexity and power consumption. Therefore, the total power consumption of a pipeline ADC increases $> N$ with a speed < 1 [11].

For low conversion speed, an SAR approach is often used since it also divides a full conversion into several comparison stages in a way similar to the pipeline ADC, except the algorithm is executed sequentially rather than in parallel as in the pipeline case. An N -bit SAR converter utilizes only one comparator with N clock cycles to complete a full conversion. Thus, the total power consumption is normalized to approximately one, while speed is now $1/N$. Since the ratio of power and speed represents the energy consumption per conversion sample, SAR converters clearly have a power efficiency advantage over the other approaches. Due to the fact that the power efficiency difference between SAR and flash topologies increases exponentially with the number of bits, N , an SAR converter provides a promising starting point of the successive approximation algorithm has traditionally been a limitation in achieving high-speed operation [11].

The ADCs for the bio-sensors or some portable devices are usually designed to be low power or energy efficient. Hence, the battery lifetime of those devices can be extended. For this issue the successive approximation A/D converter is the best choice, and there are some advantages of the SAR ADC, such as reducing the power consumption and the parasitic capacitor, with the process scale down.

1.3 Thesis Organization

The remainder of the thesis is organized as follows:

Chapter 2 describes the principle, architecture and operation of a successive approximation A/D converter. Besides, the capacitive digital-to-analog converter and latched comparator, the important building blocks of a successive approximation A/D converter, will be introduced.

Chapter 3 describes the implementation and its corresponding simulation results of an 8-bit 27 MS/s low power successive approximation A/D converter. To achieve high speed operations, a slight modification on switches' design which can achieve uniform and smaller RC time constant than conventional successive approximation A/D converter is presented here. In addition, a self-timing control mechanism which can effectively reduce the power consumption of the comparator is illustrated in this chapter.

Chapter 4 introduces an energy efficient passive charge-sharing SAR ADC first. And then, a passive charge-sharing SAR ADC is proposed and illustrated. The proposed passive charge-sharing SAR ADC has the advantages, which will be addressed in this chapter, of reducing large input capacitor and consequently saving the energy during conversion. Also, some simulation results will be shown to demonstrate the effectiveness of the proposed technique.

Finally, generalized conclusions are presented in chapter 5, and future works of this research are summarized for further exploration.

Chapter 2

The Principle and Architectures of Successive Approximation A/D Converters

2.1 Principle of Successive Approximation Algorithm

Successive-approximation A/D converters are one of the most popular approaches for realizing A/D converters due to their reasonably quick conversion time. To understand the basic operation of successive-approximation converters, knowledge of the search algorithm referred to as a “binary search” is helpful. As an example of a binary search, consider the game of guessing a random number from 1 to 128 where one can ask only question that have a “yes/no” response. The first question might be, “Is the number greater than 64?” If the answer is yes, then the second question asks whether the number is greater than 96. However, if the first answer is no, then the second question asks whether the number is greater than 32 [7].

The third question divides the search space in two once again and the process is repeated until the random number is determined. In general, a binary search divides the search space in two each time, and the desired data can be found in N steps for a

set of organized data of size 2^N [7].

Successive-approximation A/D converter applies a binary search algorithm to determine the closest digital word to match an input signal. Specifically, in the first period, after possibly the reset period, the MSB, b_1 , is determined. In the second period, the next bit, b_2 , is determined, followed by b_3 and so on until all N bits are determined. Thus, in its most straightforward implementation, a successive-approximation converter requires N clock cycles to complete an N -bit conversion. A flow graph for a signed conversion using a successive-approximation approach is shown in Figure 2-1 with $\pm 0.5 V_{\text{ref}}$ [7].

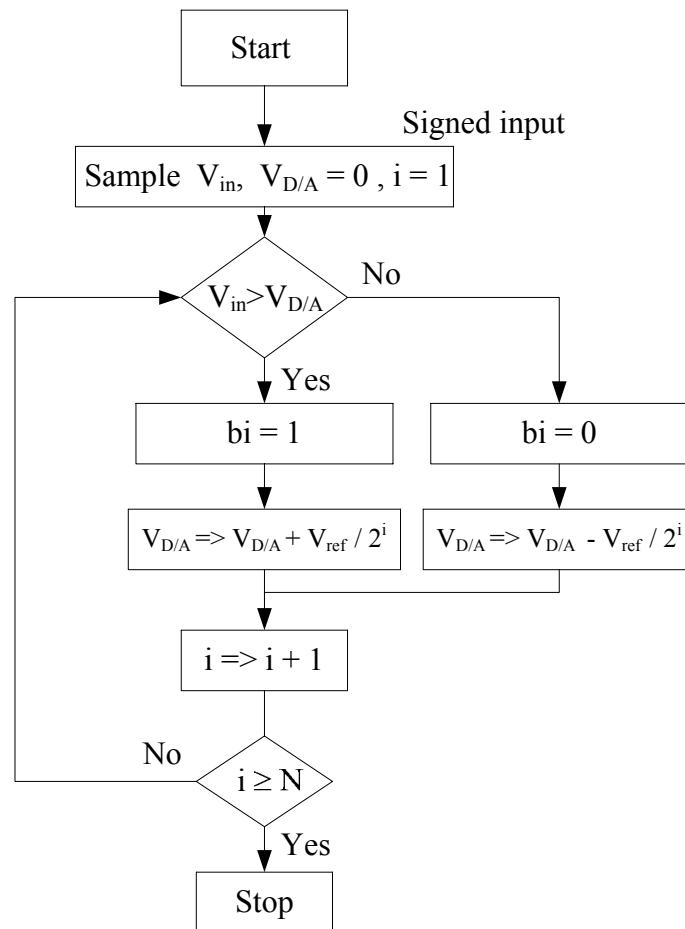


Figure 2-1 Flow graph for the successive-approximation approach [7]

2.2 The Architecture of Successive Approximation A/D Converter

2.2.1 DAC-Based Successive Approximation

The block diagram for a unipolar successive-approximation A/D converter that uses a DAC is shown in Figure 2-2. The successive-approximation register (SAR) and control logic are entirely digital and perform the necessary binary search. At the end of the conversion, the digital value in the SAR results in the voltage $V_{D/A}$ being within $0.5 V_{LSB}$ of the input signal. With this type of architecture, the D/A converter typically determines the accuracy and speed of the A/D converter. Note that a sample-and-hold is required at the input so that the value to be converted does not change during the conversion time [7].

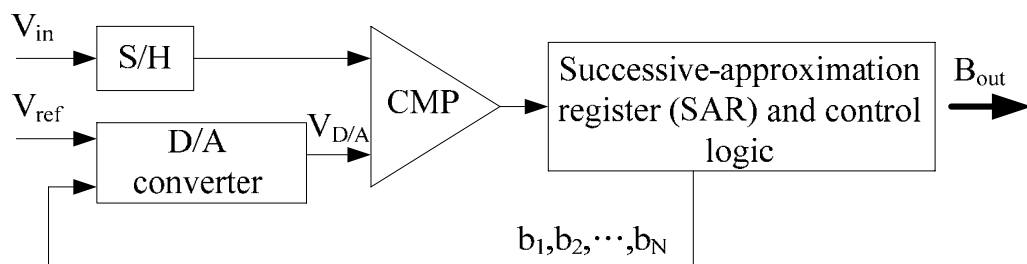


Figure 2-2 D/A converter-based successive-approximation converter [7]

2.2.2 Unipolar Charge-Redistribution A/D

The straightforward approach is to use a separate D/A converter and set it as the flow graph shown in Figure 2-3. Here, the error signal V equals the difference

between the input signal, V_{in} , and the D/A output, $V_{D/A}$. As a result, V is always compared to ground, as seen at the top of the flow graph, and the goal is to set this error difference with one LSB of zero [7].

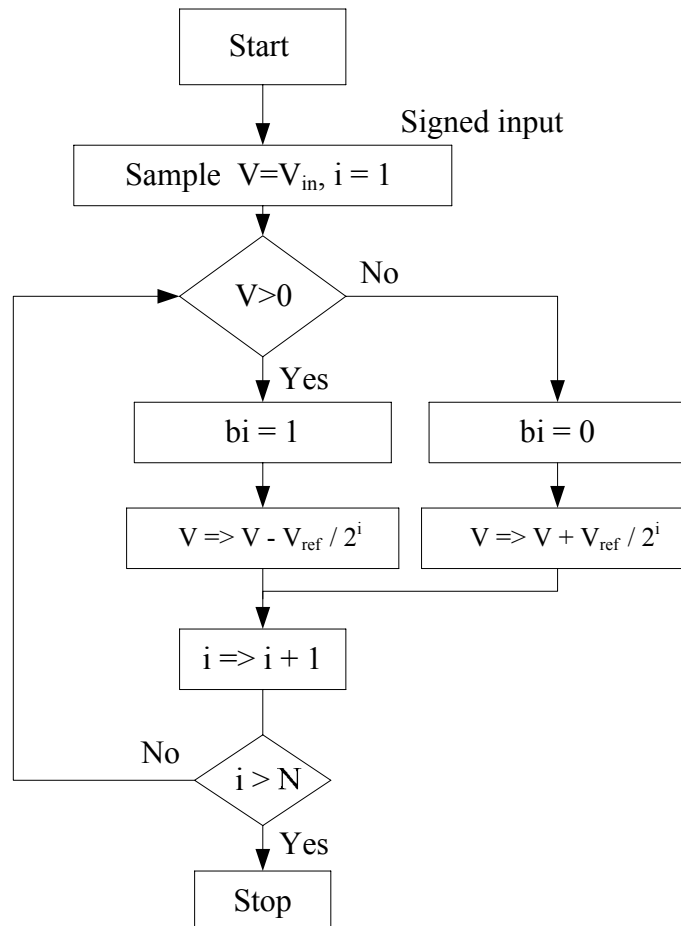


Figure 2-3 Flow graph for a modified successive approximation [7]

One of the first switched-capacitor analog systems using this approach is a charge-redistribution MOS A/D converter [5]. With such a converter, the sample-and-hold, D/A converter, and the difference portion of the comparator are all combined into a single circuit. The unipolar case is shown in Figure 2-4 and operates as follows [7]:

1. Sample mode: In the first step, all the capacitor are charged to V_{in} while the comparator is being reset to its threshold voltage through S_2 . In this step, note that the capacitor array is performing the sample-and-hold operation [7].
2. Hold mode: Next, the comparator is taken out of reset by opening S_2 , and then all the capacitors are switched to ground. This causes V_x , which was originally zero, to change to $-V_{in}$, thereby holding the input signal, V_{in} , on the capacitor array. (This step is sometimes merged with the first bit time during bit cycling). Finally, S_1 is switched so that V_{ref} can be applied to the capacitor array during bit cycling [7].
3. Bit cycling: Next, the largest capacitor (i.e., the $16C$ capacitor in this example) is switched to V_{ref} . V_x now goes to $(-V_{in} + V_{ref} / 2)$. If V_x is negative, then V_{in} is greater than $V_{ref}/2$, and the MSB capacitor is left connected to V_{ref} . Also b_1 is considered to be a 1. Otherwise, the MSB capacitor is reconnected to ground and b_1 is taken to be 0. This process is repeated N times, with a smaller capacitor being switched each time, until the conversion is finished [7].

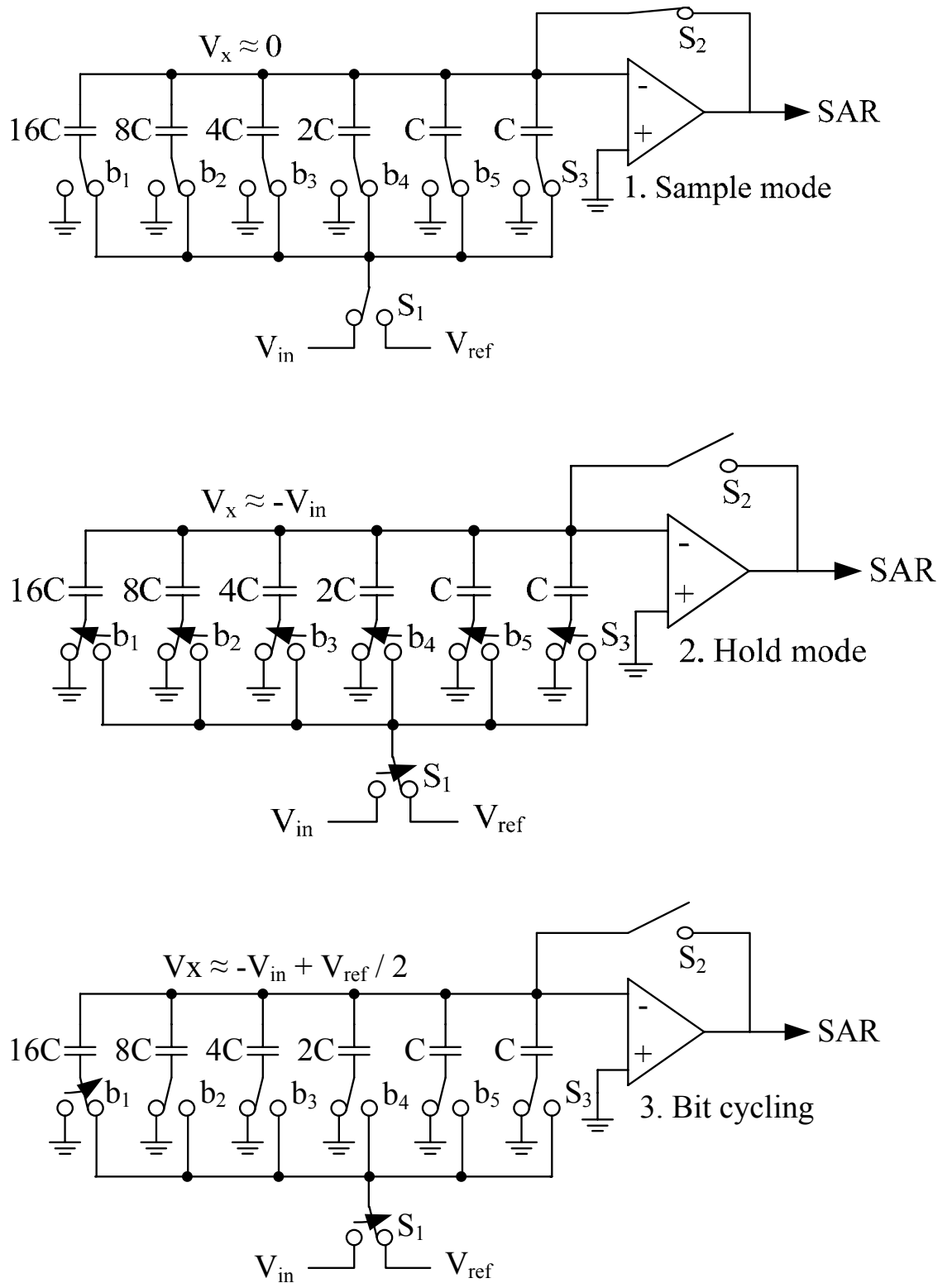


Figure 2-4 A 5-bit unipolar charge-redistribution A/D converter [7]

To get an exact division by two, note that an additional unit capacitor of size C has been added so that the total capacitance is $2^N C$ rather than $(2^N - 1)C$. Also, the capacitor bottom plates should be connected to the V_{ref} side, not to the comparator side, to minimize the parasitic capacitance at node V_x . Although parasitic capacitance at V_x does not cause any conversion errors with an ideal comparator, it does attenuate the voltage V_x [7].

A signed A/D conversion can be realized by adding a $-V_{\text{ref}}$ input. If V_x is less than zero at the first step, then proceed as in the unipolar case using V_{ref} . Otherwise, if V_x is greater than zero, use $-V_{\text{ref}}$ and test for V_x greater than zero when deciding whether to leave the capacitor connected to $-V_{\text{ref}}$ or not at each bit cycling [7].

2.2.3 Signed Charge-Redistribution A/D with a Single Reference Voltage

The same structure as the unipolar case can be used to realize a signed A/D conversion while using only a single V_{ref} if a slightly modified switching arrangement is used. Referring to Figure 2-5, and assuming V_{in} is between $\pm V_{\text{ref}}/2$, the conversion proceeds as follows [7]:

1. Sample mode: In the first step, all the capacitor, except for the largest capacitor, are charged to V_{in} while the comparator is being reset to its threshold voltage. For the signed case, the largest capacitor is now connected to $V_{\text{ref}}/2$ [7].
2. Hold mode: Next, the comparator is first taken out of reset, and then all the capacitors, except the largest one, are switched to ground. This causes V_x , which was originally zero, to change to $-V_{\text{in}}/2$. At the end of this step, the

sign of the input signal is determined by looking at the comparator output [7].

3. Bit cycling: Next, the largest capacitor (i.e., the $16C$ capacitor in this example) is switched to ground if, and only if, V_x is larger than zero (i.e., when V_{in} is less than zero). Specifically, if V_x is less than zero, then V_{in} is positive and b_1 is set to 1, and conversion proceeds as in the unipolar case, starting with b_2 , until conversion is completed. However, if V_x is larger than zero, b_1 is set to 0, the largest capacitor is switched to ground, causing V_x to become $-V_{in}/2 - V_{ref}/4$ (which is a negative value), and conversion proceeds as in the unipolar case, starting with b_2 . Once conversion is completed, some digital recoding may be required to obtain the desired output code [7].

This approach for realizing signed A/Ds has the disadvantage that V_{in} has been attenuated by a factor of two, which makes noise more of a problem for high-resolution A/Ds. Also, any error in the MSB capacitor now causes both an offset and a sign-dependent gain error. The latter causes integral nonlinearity errors [7].

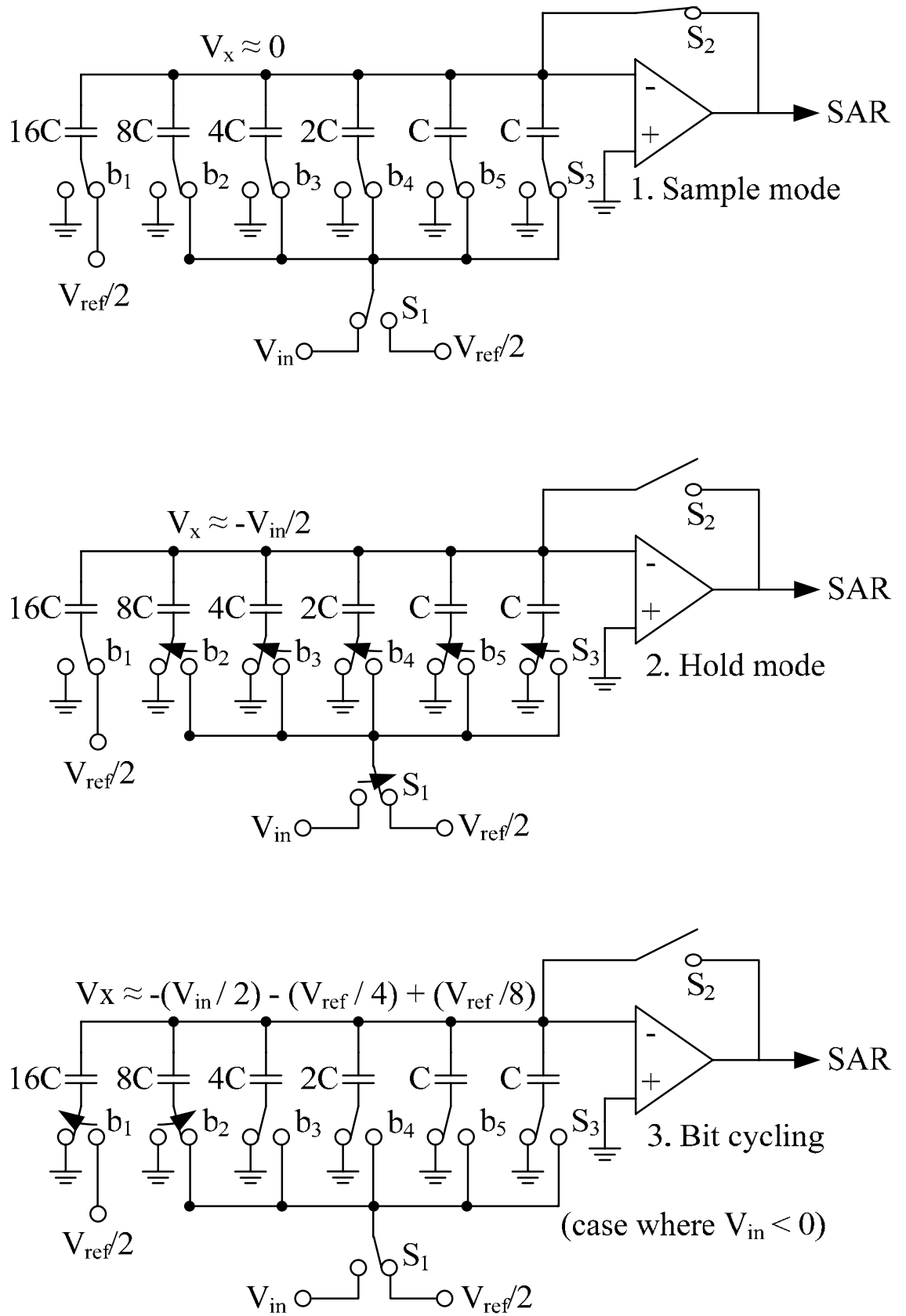


Figure 2-5 A 5-bit signed charge-redistribution A/D converter

2.3 Capacitive Digital-to-Analog Converter

The D/A converter typically determines the accuracy and speed of the successive approximation A/D converter. Usually, there are two kinds of capacitive DACs: C-2C and binary-weighted capacitor arrays, be used in the SAR ADC.

2.3.1 Binary-Weighted Capacitor Array

The use of a binary-weighted capacitor (BWC) array was first reported in 1975, as shown in Figure 2-4 [5]. Since the total capacitance value of BWC arrays rise exponentially with the resolution, the accuracy of BWC DACs is limited to 8 to 10 bit due to the large capacitor ratio and the small area to perimeter ratio of the unit capacitor in the BWC arrays [8].

To overcome these difficulties, a two-stage weighted capacitor (TWC) array was proposed in [12], where the capacitor array consists of two BWC arrays connected with a coupling capacitor, as shown in the Figure 2-6 [6].

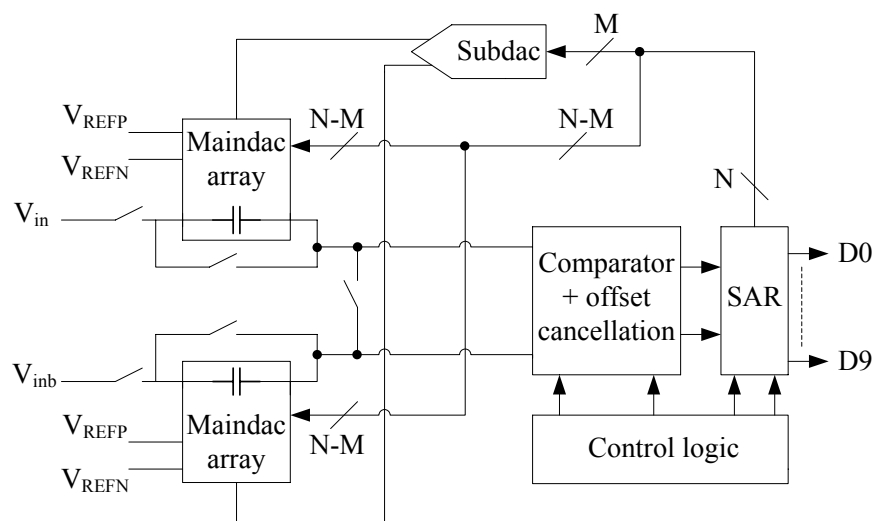


Figure 2-6 Fully differential binary weighted capacitor array [6]

The complete extension of a TWC array is a C-2C capacitor array [8]. The gain error and the nonlinearity of the BWC array is better than the C-2C capacitor array, however, it has heavy input capacitance and large capacitor ratio, so it is not suitable for high speed application.

2.3.2 C-2C Capacitor Array

An ideal C-2C capacitor array is shown in the Figure 2-7. Only two kinds of capacitor sizes, C and $2C$, are used in the capacitor array. Therefore, the C-2C capacitor array can be applied to high speed conversion and low power consumption. However, for some non-ideal effects, the C-2C capacitor array has serious problems in the gain error and linearity.

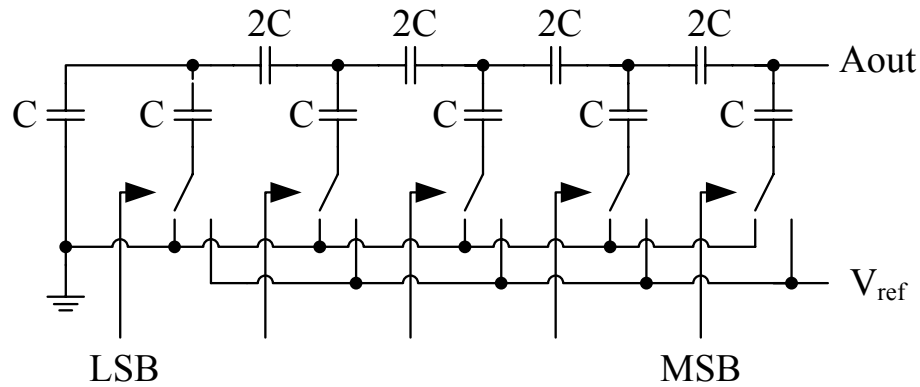


Figure 2-7 Ideal C-2C capacitor array [8]

When the parasitic capacitor C_p , including the switch, inter-routing, the top plate and the bottom plate parasitic capacitances, is considered into the ideal C-2C model, a real C-2C capacitor array is shown in the Figure 2-8.

Assuming all of the parasitic capacitors are equal to 0, for this ideal condition the capacitor array is the same as an ideal C-2C array and the C_x is equal to C . So, when

the switch MSB is connected to V_{ref} and the others are connected to ground, the voltage at the node Aout is $1/2V_{ref}$.

However, the parasitic capacitors are not equal to 0 practically, and the C_X is equal to (1). From (1), when the switch MSB is connected to V_{ref} and the others are connected to ground, the voltage at the node Aout would not equal to $1/2V_{ref}$, and it induces serious gain errors and non-linearity. Therefore, the C-2C capacitor array could not be applied to high accuracy ADCs.

$$C_x = (((((2C + 2C_{p1}) // 2C + 2C_{p2} + C) // 2C + 2C_{p3} + C) // 2C + 2C_{p4} + C) // 2C + C + C_{pvin} \dots (1)$$

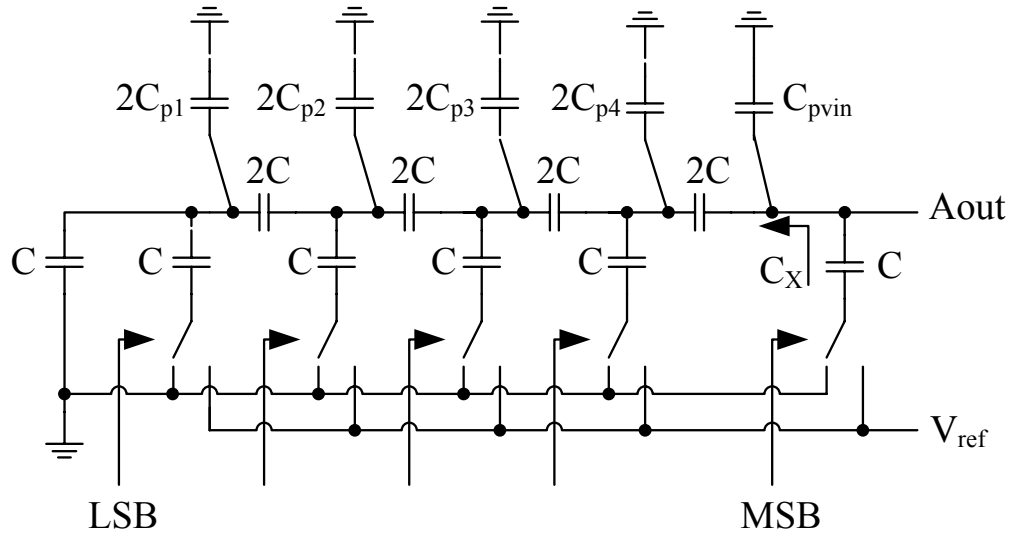


Figure 2-8 Real C-2C capacitor array [8]

2.3.3 Pseudo C-2C Capacitor Array

To solve this problem, a pseudo C-2C capacitor array model, is shown in Figure 2-9. The basic idea is to replace the C-2C capacitor array with a C-2 α C capacitor array, and adjust the capacitor ratio α in the array so that the parasitic effects are balanced out and precise voltage division are achieved [8].

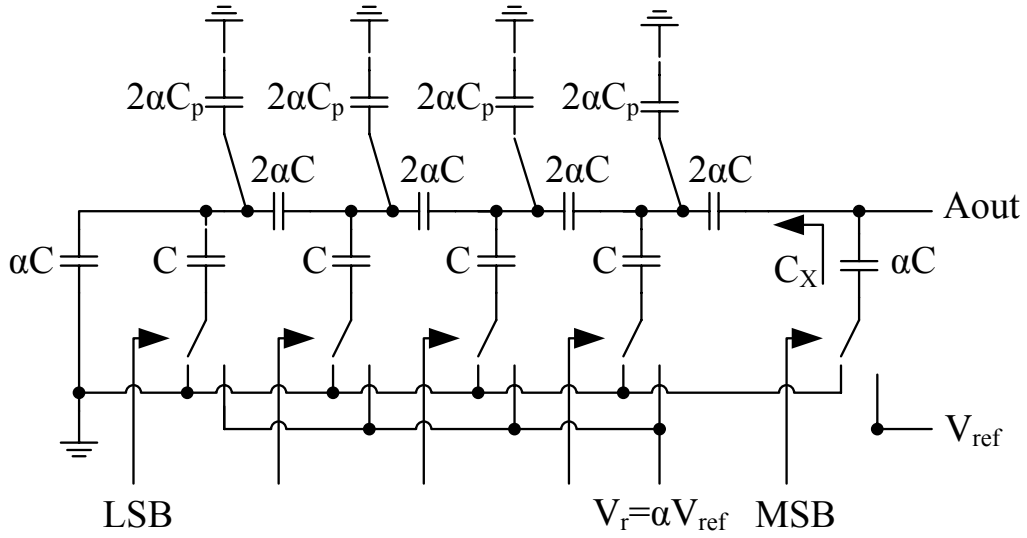


Figure 2-9 Pseudo C-2C capacitor array [8]

Directly apply the Thevenin Theorem to translate the pseudo array with parasitic capacitors in Figure 2-9 into the equivalent array structure of Figure 2-10. Each branch capacitor C' in the equivalent array combines the corresponding branch capacitance C in the C - $2\alpha C$ array with the parasitic capacitance $2\alpha C_p$ feeding in the same node

$$C' = C + 2\alpha C_p = (1 + 2\gamma\alpha)C \quad (2)$$

,where $\gamma = C_p/C$ is the ratio of bottom-plate parasitic to top-bottom inter-plate capacitance. Therefore, the choice

$$\alpha = \frac{1}{1 - 2\gamma} \quad (3)$$

Renders $C' = \alpha C$, yielding an equivalent C-2C array where all capacitances are uniformly scaled by the factor α . Since the voltage division is determined by the capacitance ratio, the charge injected to the output common node is independent of α , and identical to that of the ideal C-2C array shown in Figure 2-7. Notice that the branch capacitors driven by the reference voltage in Figure 2-9 are C rather than $C' = \alpha C$ in Figure 2-10, so the reference voltage in the pseudo array is scaled by α

[8].

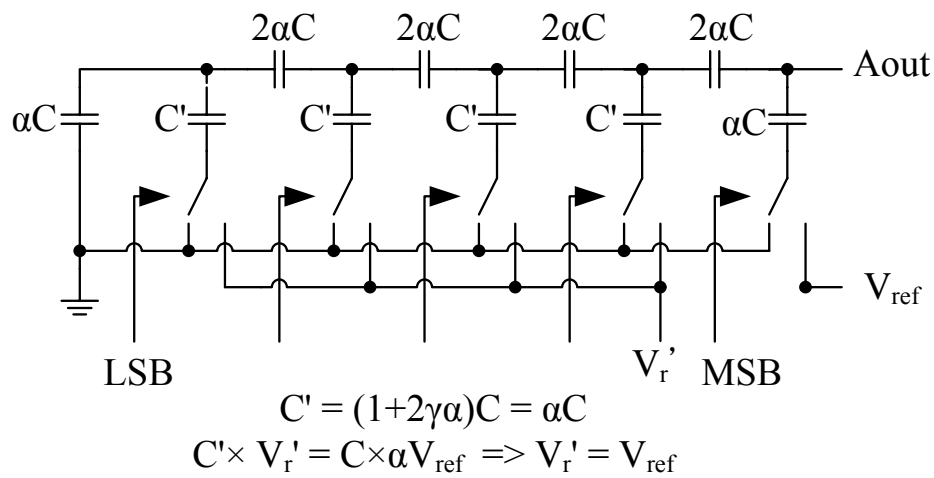


Figure 2-10 Thevenin equivalent of pseudo C-2C capacitor array [8]

2.4 Latched Comparator

Comparator is an important component in the successive approximation A/D converter, and the latched comparator is usually used. If a latch toggles in one direction, it might have a tendency to stay in that direction, therefore a reset mechanism is used to remove this Hysteresis effect. For the reason, the operation of the latched comparator is divided into two parts, the reset and regeneration phase. However, the reset mechanism would cause the kickback noise and affect the input signal of the comparator.

There are three kinds of the latched comparator, the static latched comparator, the class-AB latched comparator and the dynamic latched comparator, are shown below [13].

2.4.1 Static Latched Comparator

A static latched comparator is represented in Figure 2-11. The regeneration of a static latched comparator is done by two class-A cross-coupled inverters. This current-limited operation leads to a slow regeneration process. The power efficiency is poor, since the consumption is purely static [13].

There is always a differential pair acting as pre-amplifier, whose output current is either mirrored or injected through a cascade transistor in the regeneration nodes. This isolation between the drains of the differential pair transistors and the regeneration nodes reduces the kickback noise, but introduces two poles: one at the intermediate node, where the drain of the differential pair connects, and the other in the regeneration nodes [13].

This latched comparator presents low kickback noise, but the static power consumption and slow regeneration process does not make it attractive [13].

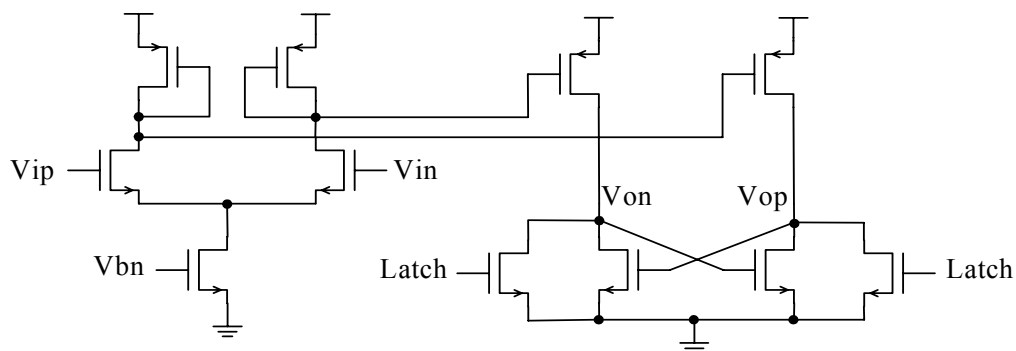


Figure 2-11 Static latched comparator [13]

2.4.2 Class-AB Latched Comparator

The regeneration of class-AB latched comparator is done by two cross-coupled CMOS inverters, as shown in Figure 2-12. Their current increases momentarily, during the regeneration process, to charge the output nodes faster—class-AB operation [13].

The drains of the input differential pair are directly connected to the regeneration nodes. The circuit reacts quicker to input variations, because there is only one pole. However, this increases the kickback noise because there are now rail-to-rail signals at nodes capacitively coupled to the inputs [13].

We can conclude that these comparators are faster and more power efficient than the static latched comparators, but generate more kickback noise [13].

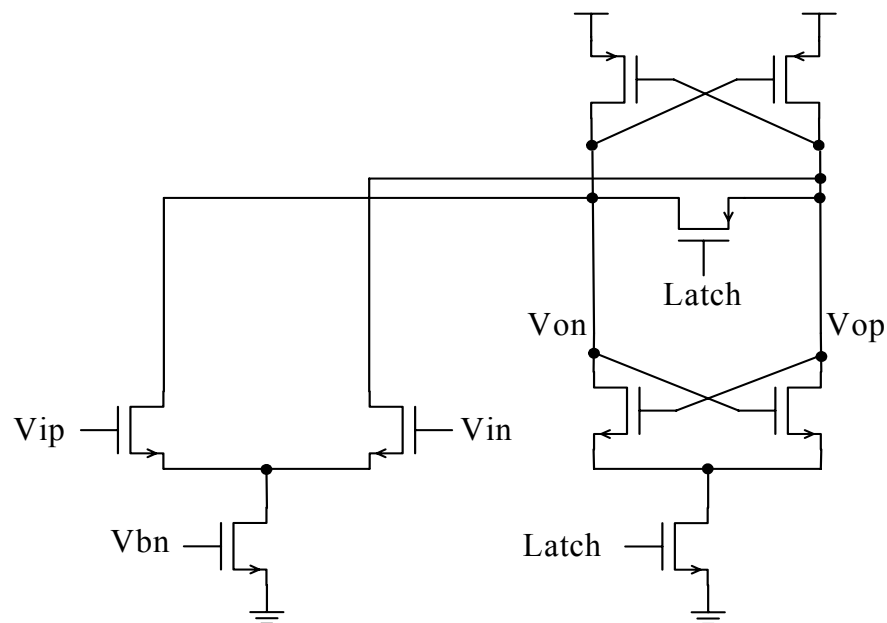


Figure 2-12 Class-AB latched comparator [13]

2.4.3 Dynamic Latched Comparator

Class-AB latched comparators, although more power efficient than their static counterparts, still have supply current in the reset phase and after the regeneration finishes. In dynamic latched comparators, the current only flows during the regeneration, as shown in Figure 2-13 [13].

After regeneration is completed, one of the output nodes is at VDD; the other output and both drains of the differential pair have a 0-V potential. There is, in this situation, no supply current, which maximizes power efficiency [13].

The nodes where the drains of the input differential pairs connect have rail-to-rail excursion, originating a large kickback noise. There is, in this type of comparators, another kickback noise source: the variation of the operating region of differential pair transistors [13].

It can be concluded that the fastest and most power efficient comparators generate more kickback noise [13].

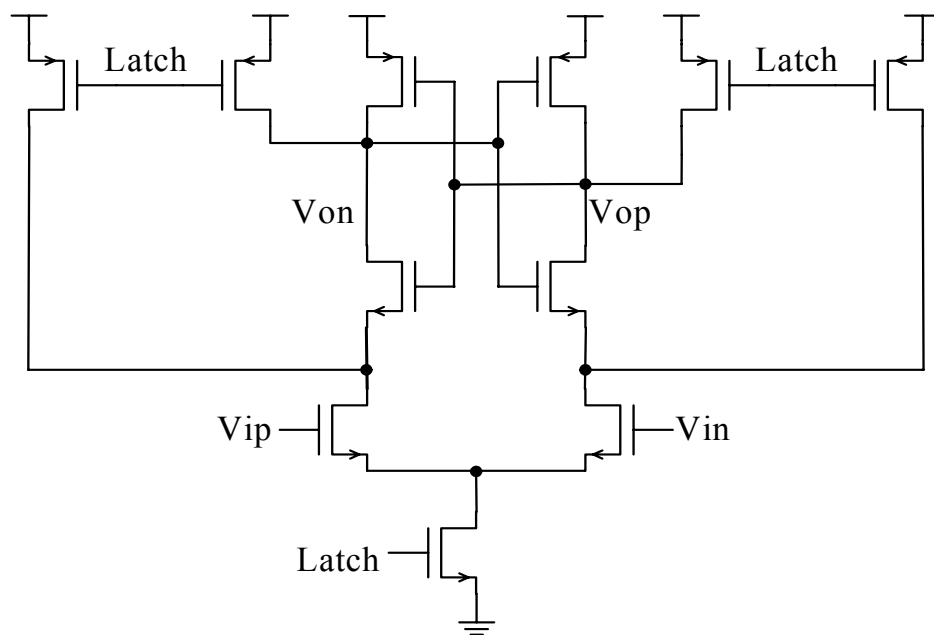


Figure 2-13 Dynamic latched comparator [13]

Chapter 3

An 8-bit 27MS/s Successive Approximation Analog-to-Digital Converter

3.1 Motivation

Digital television (DTV) is converts the television signal to digital signal and then broadcasts it after the especial digital signal process (DSP). The digital signal would be converted to television signal and displayed on the screen after receiving it, as shown in the Figure 3-1.

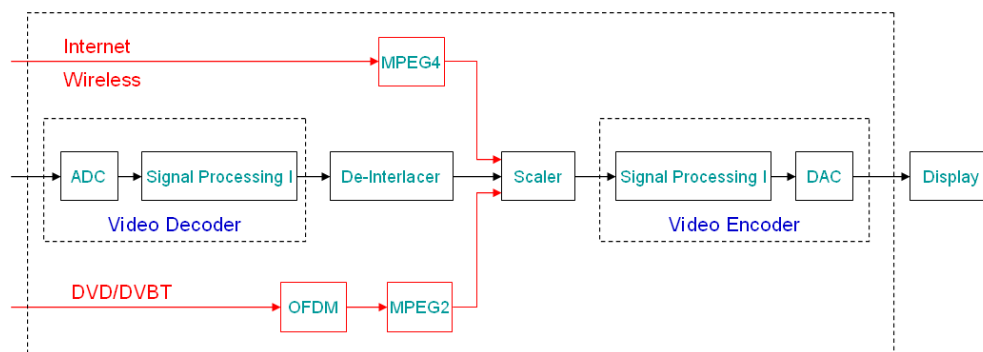


Figure 3-1 Digital television architecture

The specifications of ADCs used in the DTV are usually need 8-bit or even

10-bit resolution, and the conversion rate is 27MS/s, 54MS/s, or 108MS/s.

An 8bit 27MS/s successive-approximation ADC is proposed to meet the requirement of DTV applications in this chapter.

3.2 The Architecture of Proposed Successive Approximation A/D Converter

Figure 3-2 shows a conventional unipolar charge-redistribution successive approximation ADC. In the sampling phase, all the capacitors in the capacitor array are charged to V_{in} , it needs a lot of time to sample the input voltage. For this reason, the conversion rate of a conventional unipolar charge-redistribution successive approximation ADC is difficult to increase, a proposed successive approximation ADC is shown in Figure 3-3.

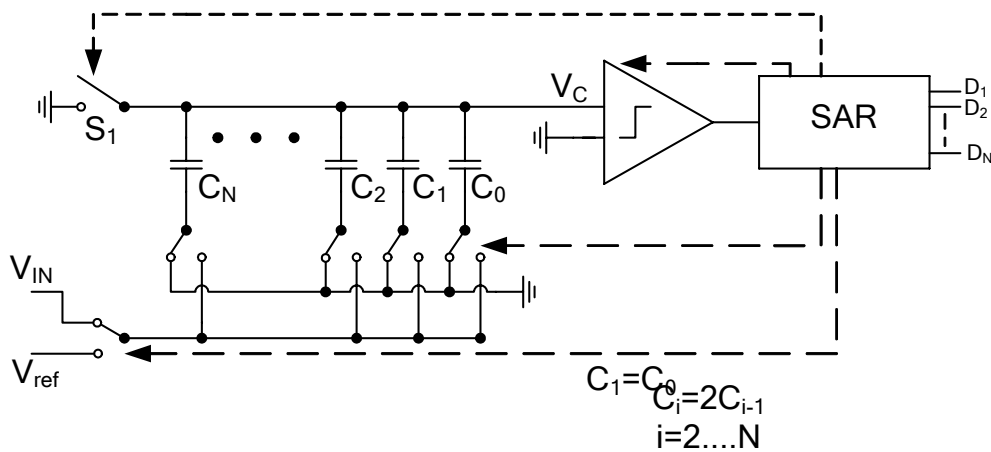
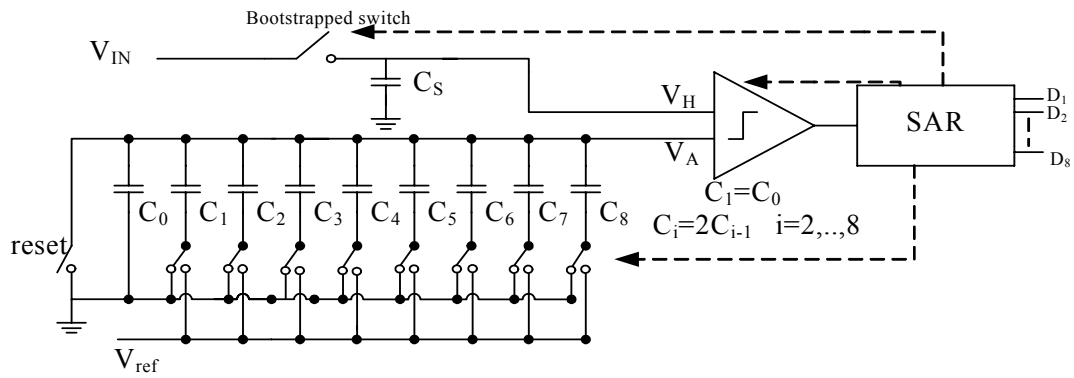


Figure 3-2 Conventional unipolar charge-redistribution SAR ADC [7]



The proposed successive approximation ADC separates the sample-and-hold and the digital-to-analog converter into two parts. The sample-and-hold, including a bootstrapped switch and a sampling capacitor, operates at the negative port of the comparator, and the digital-to-analog converter, including several ground switches and a binary weight capacitor array operates at the positive port of the comparator [10]. Therefore, in the sampling phase only the sampling capacitor must be charged to V_{in} , it decreases the sampling time.

The proposed SAR ADC consists of a sample-and-hold (S/H) circuit, a charge-redistribution digital-to-analog converter (DAC), a comparator, and a successive approximation register (SAR). With the proposed comparator which has a rail-to-rail input common-mode range, the input range of the ADC is also rail-to-rail. Most of the switches in the proposed SAR ADC are grounded switches and only one bootstrapped switch is required. Thus saves the total power consumption [10].

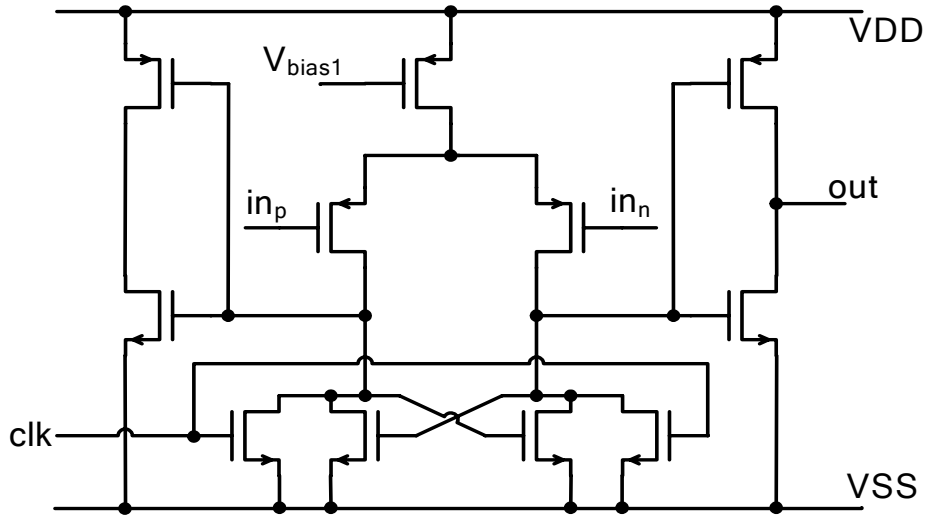


Figure 3-5 Conventional latched comparators with one differential pair

The proposed comparator is shown in Figure 3-6. Both differential pairs convert the differential input voltage into differential output currents. These currents, which formed by N_4 and N_6 , will sum together to drive the regenerative load. Two inverters with a specified transition voltage amplify the complementary outputs to full rail-to-rail logic levels. Additional resetting NMOS N_3 and N_5 are used to reduce the hysteresis effect and speeds up the comparison [10].

In order to decrease the power consumption, the bias1 is connected to reset and the bias2 is connected to reset_b, the inverting function of reset. With this design, there is no current pass through the PMOS P_4 and the NMOS N_2 during the reset phase of the comparator, as shown in Figure 3-7. An XOR gate is used at the comparator output node to generate a Valid signal, which can detect whether the regeneration phase be completed or not. When the Valid signal is triggered to high, then the reset signal will be set to VDD again, as shown in Figure 3-8. With this self timing controller, the operating time of the comparator in regeneration phase will be decreased, hence the power consumption of the comparator can be saved

approximately 50% [14].

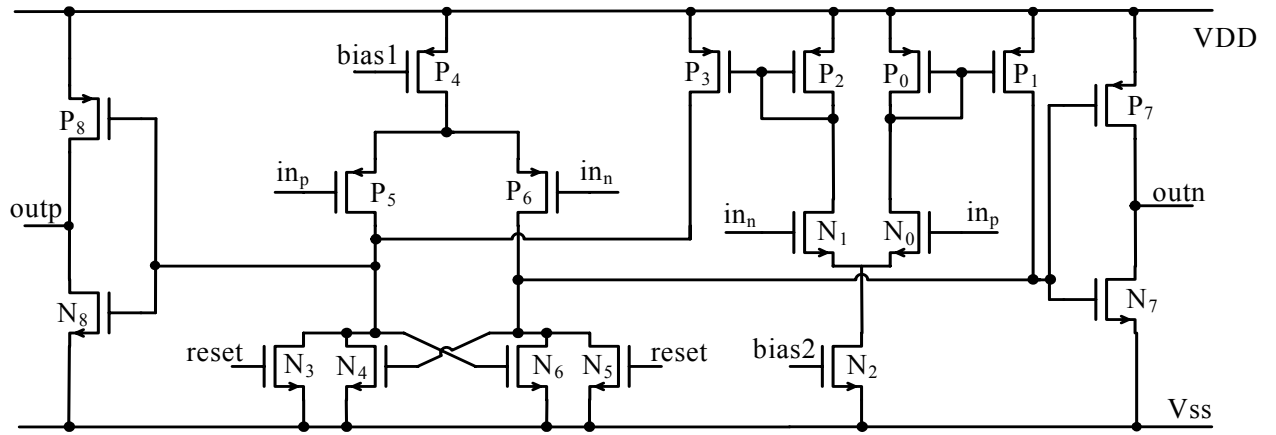


Figure 3-6 Proposed comparator with a rail-to-rail common mode input range [10]

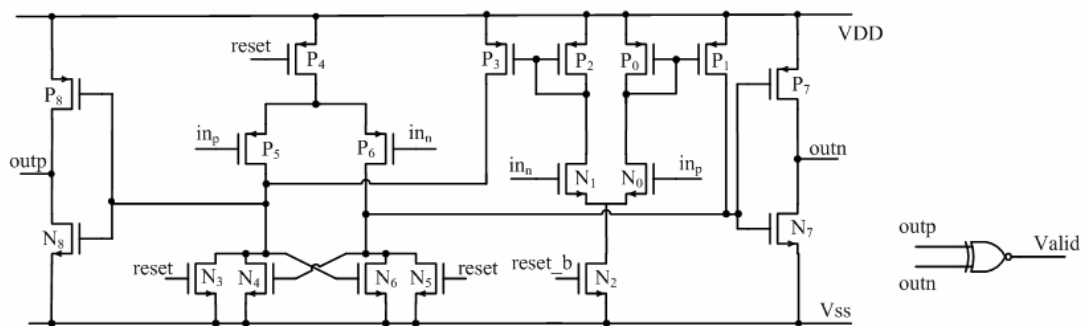


Figure 3-7 Proposed comparator with self timing controller

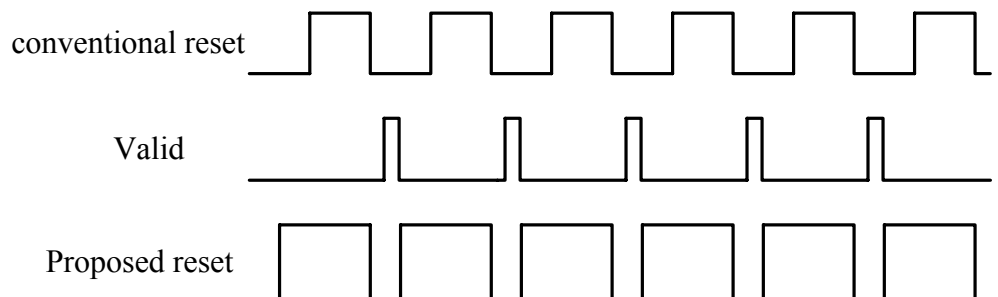


Figure 3-8 Control waveforms of self timing controller

There is no offset cancellation mechanism in the ADC due to low power consideration [10].

3.2.3 Digital-to-Analog Converter (DAC)

The DAC is implemented using a binary-weighted switched-capacitor array as shown in Figure 3-9. S_1 to S_8 are the control signals generated by the SAR to control the multiplexers of the DAC. Thanks to the ADC architecture, the switch connecting to V_{ref} are implemented by a simple PMOS and a simple NMOS as the switch connecting to ground V_{ss} [10].

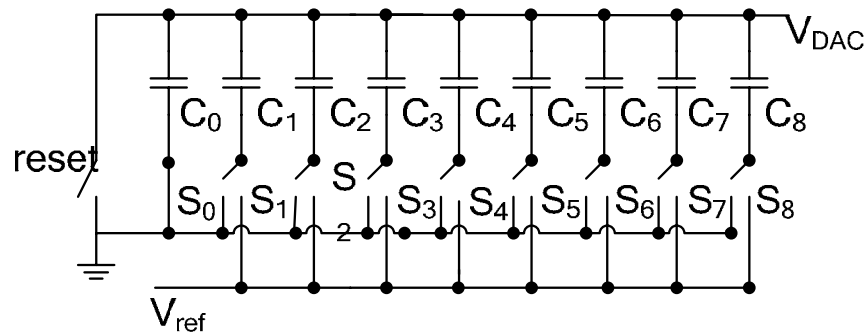


Figure 3-9 Binary-weighted switched-capacitor array [10].

The total capacitance in the DAC port should be kept smaller to save the power dissipation and increase the conversion rate. In practice, the smallest capacitance of the unit capacitor is determined with respect to the thermal noise, capacitor matching, and the design rules. For 8-bit resolution, the unit capacitance is not limited by the KT/C thermal noise. Consequently, the unit capacitance is set to the minimum value according to the design rules which is about 25 fF [10].

The major limitation on speed with charge redistribution converters is often due to the RC time constants of the capacitors and switches. In Figure 3-10, a

conventional RC time constant model of the capacitor array is shown. The open-circuit time constant approach [15] can be used to obtain an estimate of the high-frequency time constant by summing individual time constants due to each capacitor [7].

For a synchronous SAR ADC, the RC time constant of the largest capacitor usually determines the clock period. In order to increase the speed, a modified RC time constant array model is shown in the Figure 3-11, in this model the largest capacitor $128C$ is connected to the smallest resistor $R/128$, which can decrease the clock period. Those resistances, $R/2$, $R/4$, ..., $R/128$, can be realized by paralleling the switches. During the conversion process, each of the capacitors would be connected to V_{ref} or ground by the ground switches, so there is no the charge injection noise. For this mechanism, it can raise the conversion rate up to 27MS/s.

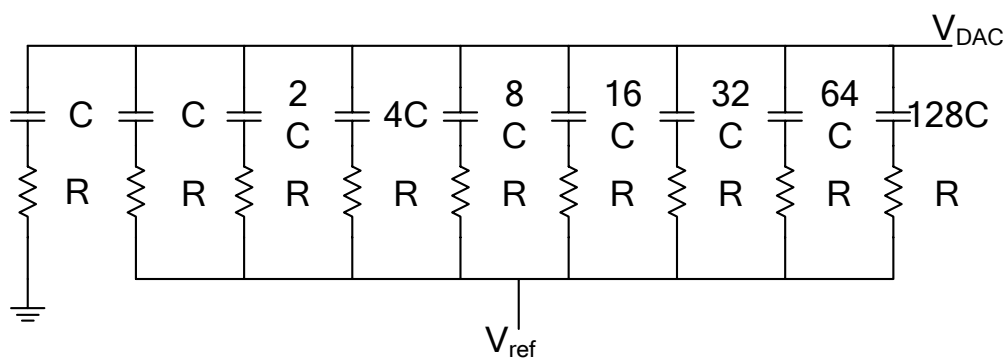


Figure 3-10 Conventional RC time constant model of capacitor array

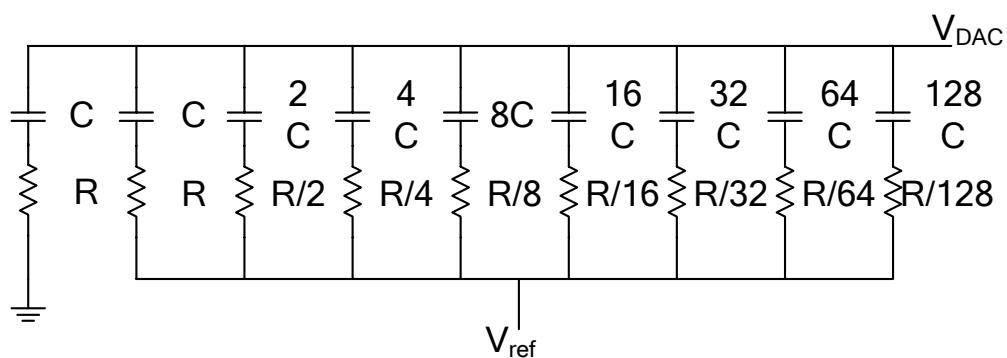


Figure 3-11 Modified RC time constant model of capacitor array

3.3 Simulation Results

The chip occupies 0.04 mm^2 core area fabricated through TSMC using $0.13\mu\text{m}$ mixed signal 1P8M 1.2V/3.3V CMOS technology. The process offers metal-insulator-metal (MIM) capacitor and many kinds of resistors.

3.3.1 Functional Simulation

The functional simulation result is shown in Figure 3-12, in which the curve V_S means the sampling voltage at the positive node of the comparator and the curve V_{DAC} means the compared voltage at the negative node of the comparator. During the successive approximation process, V_{DAC} will approximate the sampling voltage V_S with binary-scaled steps. Finally, the different voltage between V_{DAC} and V_S will be less than 1 LSB.

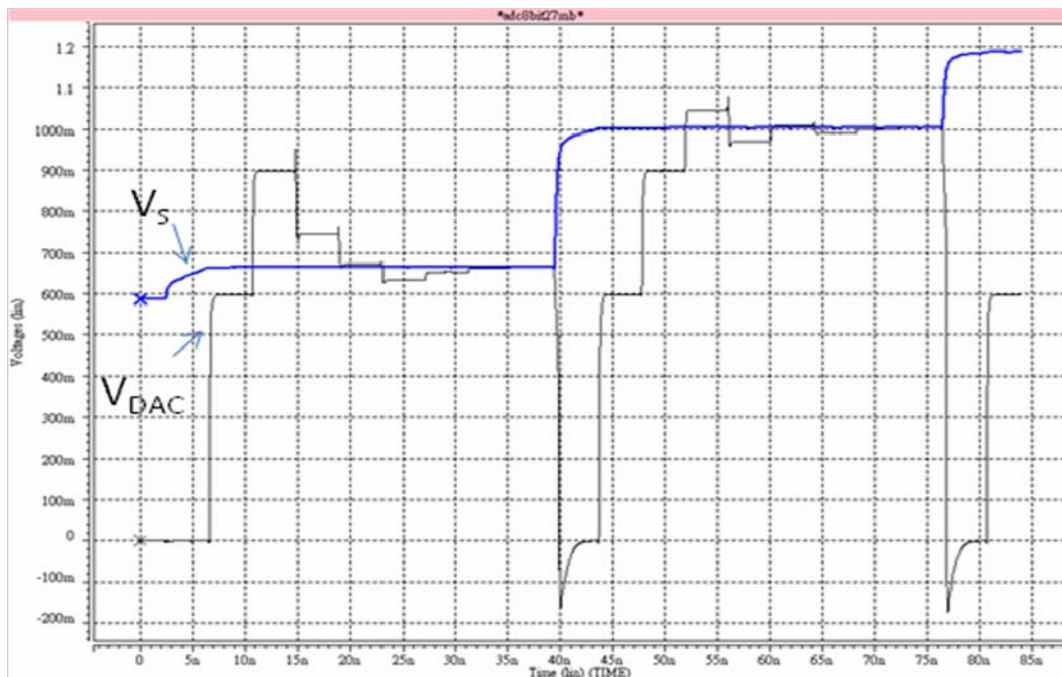


Figure 3-12 Functional simulation result

3.3.2 Pre-Layout Simulation Results

This 8-bit 27MS/s SAR ADC uses the TSMC 0.13 μ m 1P8M technology. For FFT analysis, 512 point are sampled with $f_{in} = 2.7$ MHz and 27MS/s conversion rate, as shown in the Figure 3-13, where the SNDR = 49.01 dB and ENOB = 7.85bit.

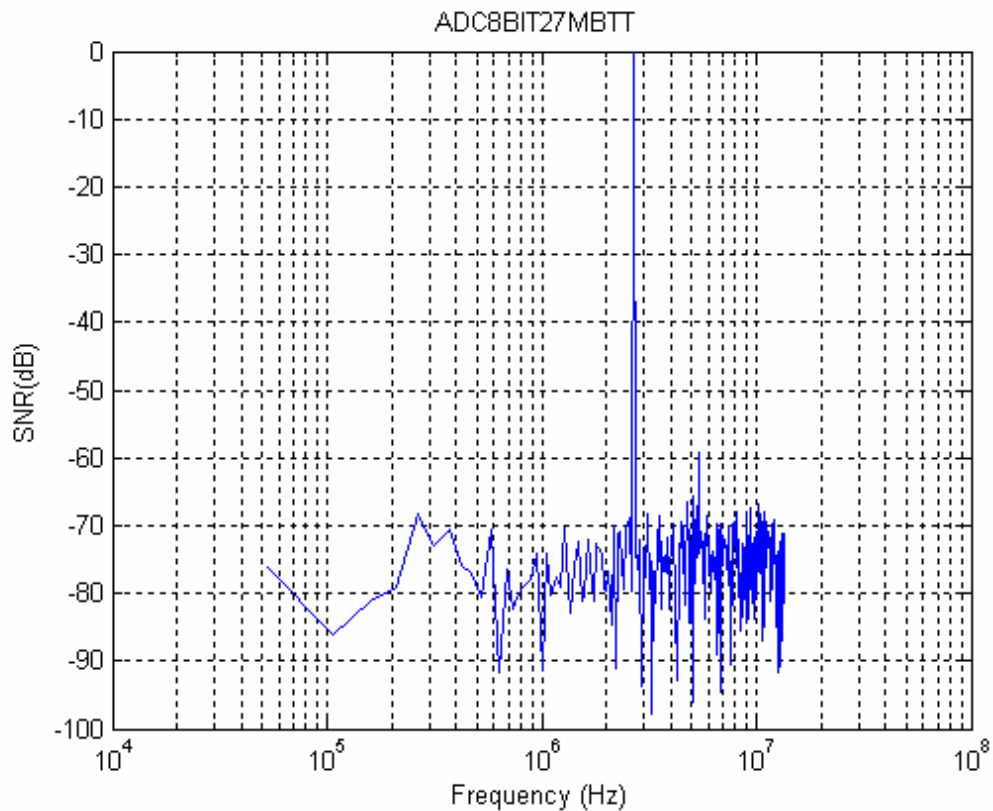


Figure 3-13 FFT simulation result of the output code with $f_{in} = 2.7$ MHz and 27MS/s conversion rate (SNDR = 49.01dB and ENOB = 7.85 bit)

Table 3-1 lists the simulation results of the proposed ADC in different process corners, Table 3-2 presents the ADC performances at different temperatures, and Table 3-3 shows the ADC performances from different supply voltages. In Table 3-1, the item FS means fast NMOS transistors and slow PMOS transistors, and the other items can be explained in the same way. The affection of the temperature variation on

the ADC performance is rather small, but this does not mean we can ignore the temperature effect. Temperature variation around the chip may degrade the performance or damage the chip. In the case of different supply voltages, the lower supply voltage may cause the switches not be fully turn-on as they do under higher supply voltage. Therefore, the ADC performance will be affected by the variation supply voltage.

Input / Corner	SNDR / ENOB
2.7MHz / TT	49.01 / 7.85
2.7MHz / FF	46.97 / 7.51
2.7MHz / SS	47.03 / 7.52
2.7MHz / FS	47.45 / 7.59
2.7MHz / SF	48.11 / 7.70

Table 3-1 Corner deviation simulation

Input / Temperature	SNDR / ENOB
2.7 MHz / 80°C	48.29 / 7.73
2.7 MHz / 0°C	49.31 / 7.90

Table 3-2 Temperature deviation simulation

Input / Supply voltage	SNDR / ENOB
2.7 MHz / 1.08V	48.41 / 7.75
2.7 MHz / 1.32V	49.14 / 7.87

Table 3-3 Supply voltage deviation simulation

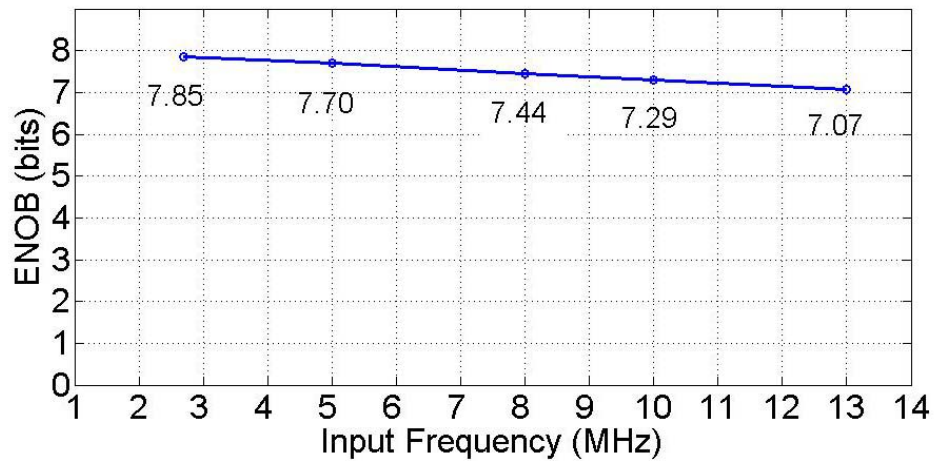


Figure 3-14 ENOB with respect to input frequency at 27MS/s

The ENOB with respect to different input frequency is shown in Figure 3-14, the ENOB still maintain near 7.07 bits near Nyquist rate at 27MSample/s.

The total power consumption of the proposed SAR ADC is 385 μ W, it composes of each component as follows: DAC 43%, comparator 34%, digital 21%, sample-and-hold 1%, as shown in Table 3-4. Most of the power is consumed at the charging and discharging of capacitor array. Half of the power consumption can be saved with the self timing controller, hence the total power consumption of the comparator is just 131 μ W.

Component	Power(μ W)
Comparator	131
S/H	6
DAC	166
SAR	82
Total	385

Table 3-4 Power consumption of each component

After all the pre-layout simulations are introduced, specifications of the proposed SAR ADC are summarized in Table 3-5. Signal-to-noise-and-distortion ratio with 2.7MHz input is 49.01dB at 27MS/s. The total core area is $0.2 \times 0.2 \text{ mm}^2$ for TSMC 0.13 μm 1P8M technology. The ENOB of all the deviations, including the process, temperature, and supply voltage deviation, are approximately 7.4~7.8 bits. The FOM of this work is just 105 fJ/conversion-step, which is very energy efficient for A/D converters.

Specifications	Results
Supply Voltage	1.2V
Sampling Frequency	27 MHz
Input Frequency	2.7 MHz
Input Common-Mode Voltage	0.6V
Input Range	1.2 V _{p-p}
Output Form	Parallel out
ENOB	7.4 ~ 7.8
Power	380 μW (Avg.)
Area (core)	$0.2 \times 0.2 \text{ (mm}^2\text{)}$
Pin Counts	16
Resolution	8 bits
Process	TSMC 0.13 μm 1P8M Process
FOM	105 fJ / Conversion-Step

Table 3-5 The ADC specification

3.3.3 Post-Layout Simulation Results

To consider the matching of capacitor array, the common centroid and dummy mechanisms are used in the layout of the capacitor array, as shown in Figure 3-15. The LSB capacitors are placed at the two edges of the capacitor arrays for the routing and layout rules considerations. The MIM capacitors, metal 7 and metal 8, are used to accomplish the capacitor arrays which have less parasitic capacitor. The layout of the capacitor array is shown in the Figure 3-16.

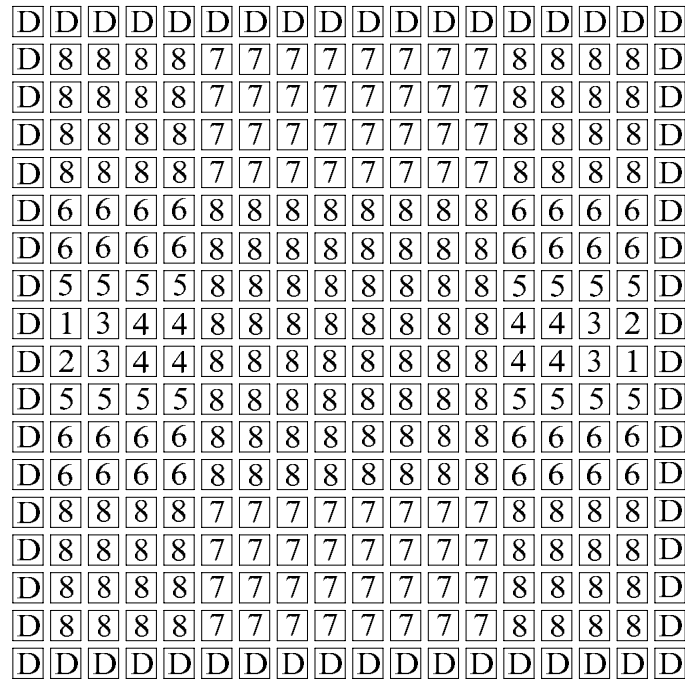


Figure 3-15 Floor plan of common centroid capacitor arrays

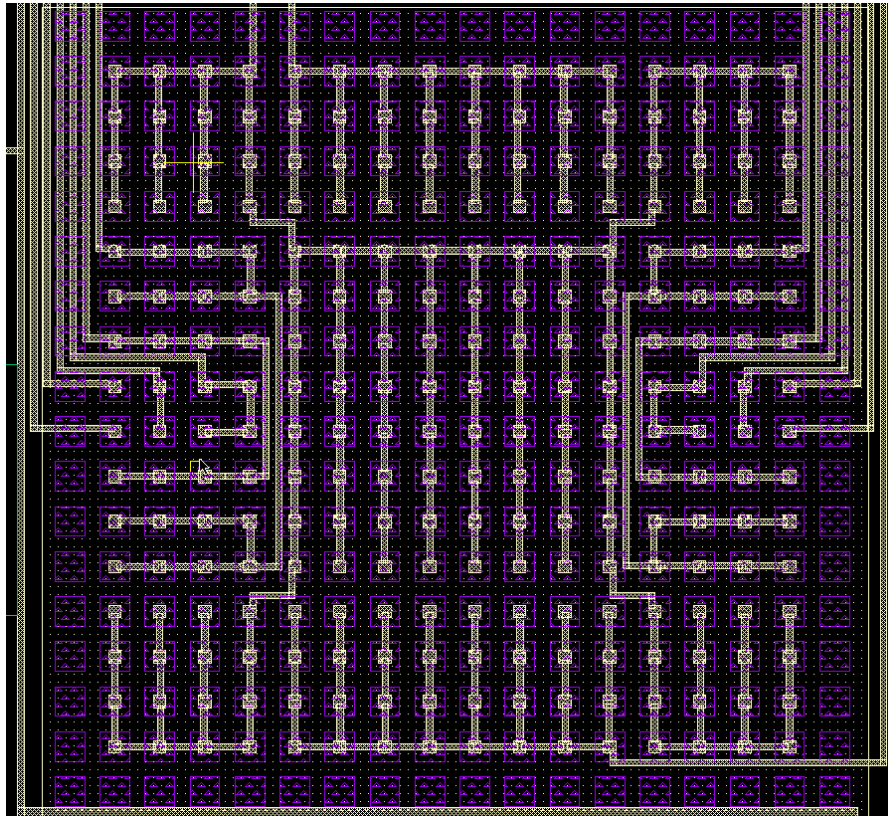


Figure 3-16 Layouts of the capacitor array

The layout of SAR and digital control circuits is shown in Figure 3-17. In order to decrease the power consumption, the static logics are used in digital circuits.

Figure 3-18 shows the layout of the sample-and-hold. It needs a sampling capacitor and a boosted clock driver.

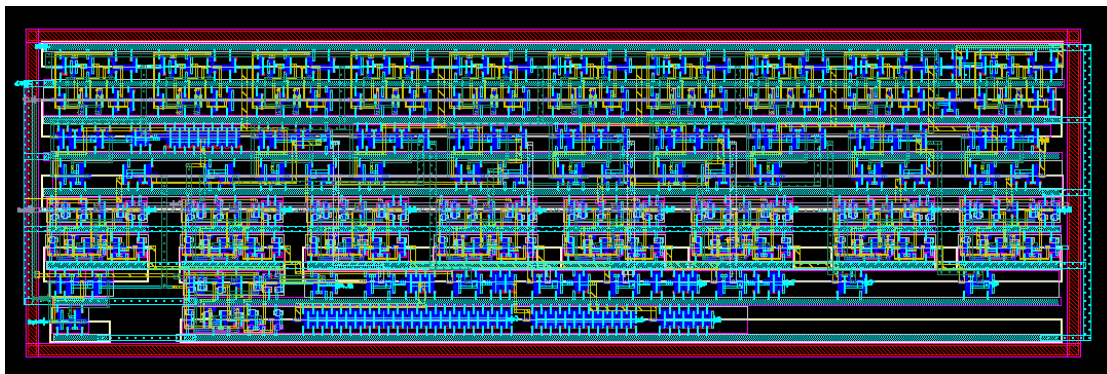


Figure 3-17 Layout of the SAR and digital control circuits

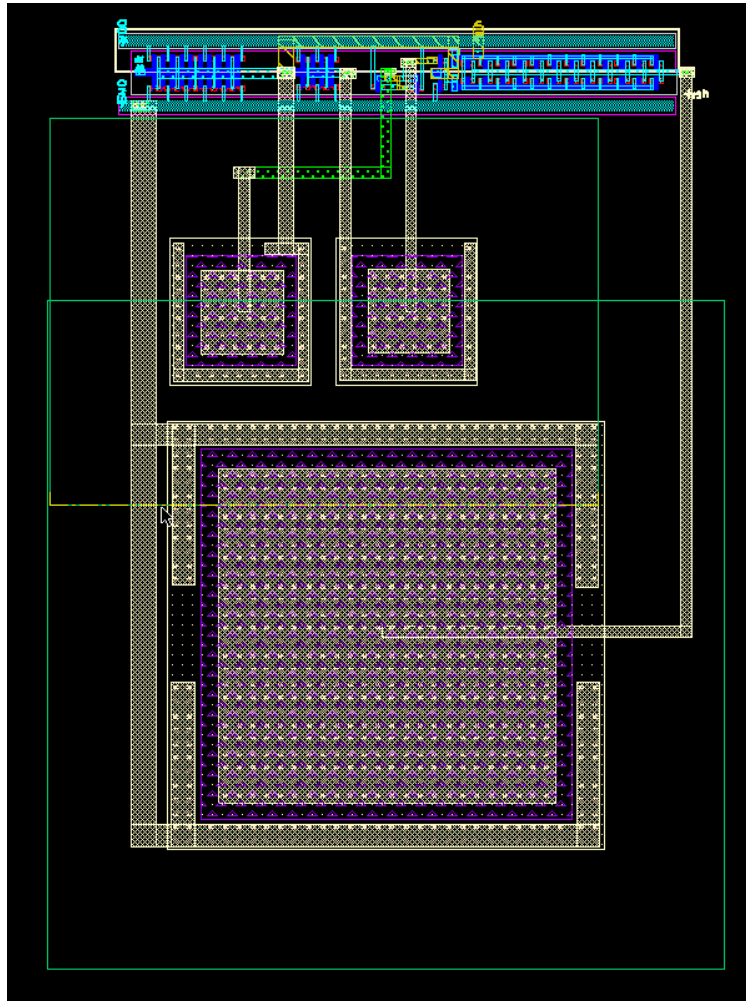


Figure 3-18 Layout of the sample and hold

After all the building blocks are laid out, we combine all the necessary circuits together. The floor-plan is depicted in Figure 3-19, and the chip layout is shown in Figure 3-20. The chip occupies 0.04 mm^2 die area and the capacitor array occupies half of the chip area. The total area of the proposed ADC is very small and economical.

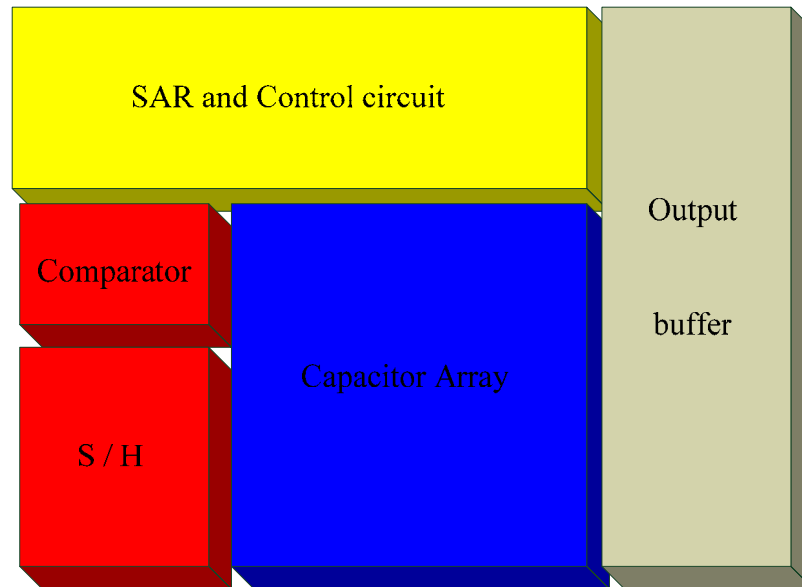


Figure 3-19 Floor-plan of the proposed ADC

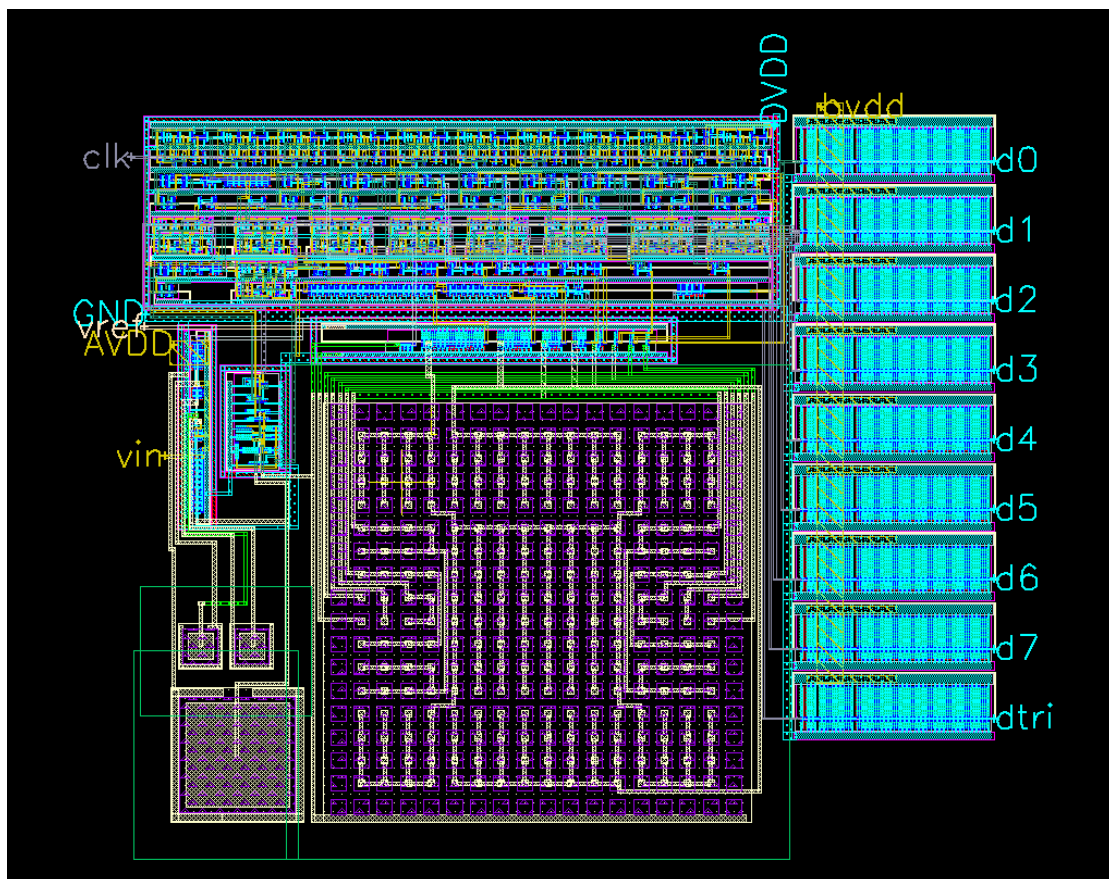


Figure 3-20 Layout of the chip (core)

The signal to noise and distortion ratio of the post-layout simulation with 2.7 MHz is 44.6dB at 27MS/s. The FFT simulation result is shown in Figure 3-21.

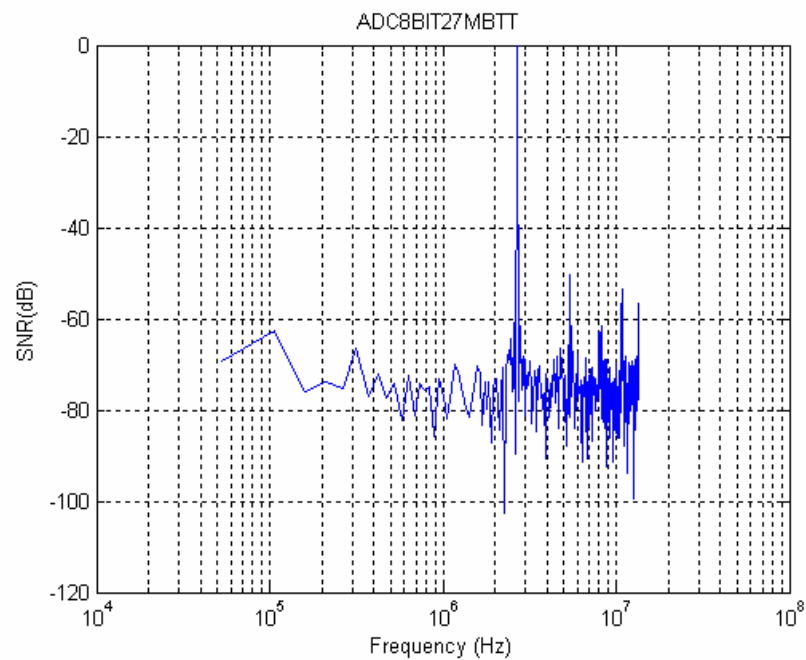


Figure 3-21 FFT simulation result of the output code with $f_{in} = 2.7\text{MHz}$ and 27MS/s conversion rate (SNDR = 44.6dB and ENOB = 7.12 bit)

The photograph of the whole chip is shown in the Figure 3-22.

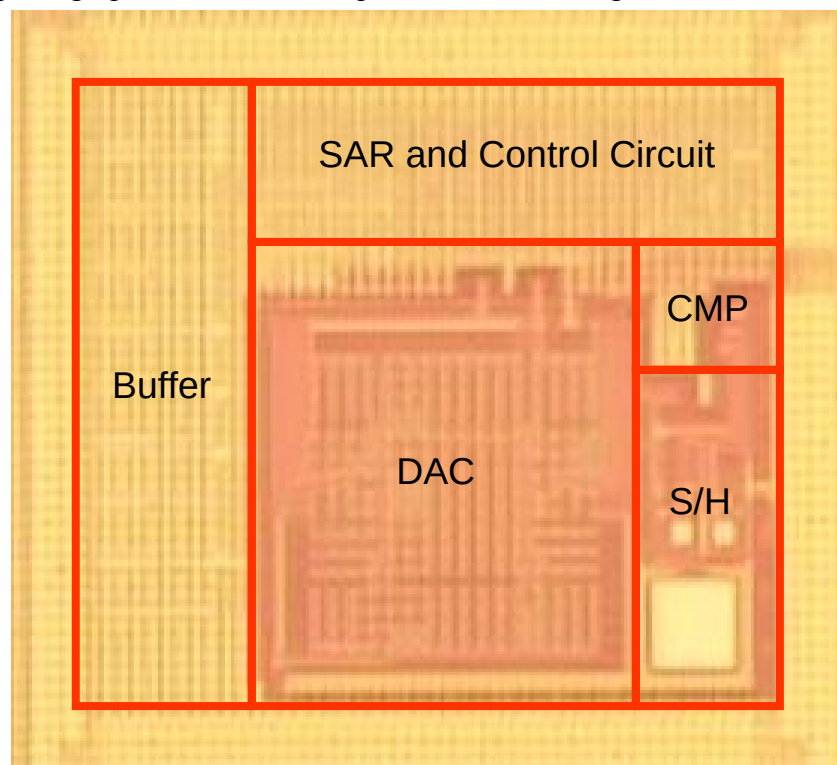


Figure 3-22 Photograph of the whole ADC ($0.2 \times 0.2 \text{ mm}^2$)

Chapter 4

Energy Efficient Passive Charge-Sharing (PCS) Successive Approximation A/D Converter

4.1 Motivation

ADCs with 8-to 10-b accuracy operating at several tens of MS/s have for a long time been the territory of pipeline architectures. But in scaled CMOS, these specifications have become well in range of SAR ADCs. Most SAR ADCs use an operating principle similar to the charge redistribution architecture [5]. This requires fast-settling opamps in both the input and the reference voltage, able to settle their output voltage in a very short time while driving large capacitive loads. Also the high-speed must be available. Using the typical FOM definition for ADCs, current state-of-the-art FOM is around 0.2pJ in [2] [3], and often this does not take the reference buffers or clock generation into account [1].

4.2 The Architectures of Conventional Passive Charge-Sharing SAR ADC

SAR architecture uses PCS (instead of active charge redistribution) to both sample the input signal and perform the binary-scaled feedback during the successive approximation. The basic architecture depicted in Figure 4-1 works completely in the charge domain. The input is sampled on a capacitor, and charge is added or subtracted until the result converges to zero. Simple passive switches, instead of any active circuits, are used to add/subtract these charges. The only active element is the comparator, which is the basic principle that allows achieving the fundamental lowest limit on power consumption [1].

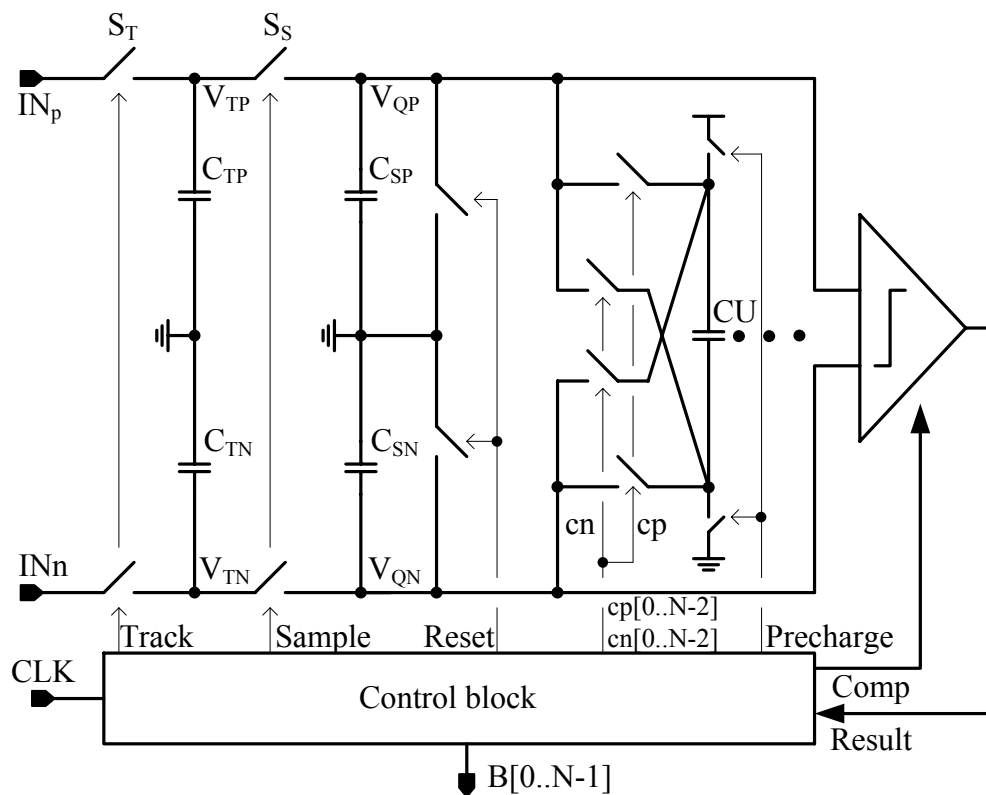


Figure 4-1 Basic charge-sharing SAR ADC architecture [1]

The operation of the ADC can be explained with the waveforms shown in Figure 4-2. Before the start of conversion, the sampling capacitors C_S are reset to zero. The tracking switch S_T is closed and the sampling switch S_S open, so the charge in the tracking capacitors C_T follows the differential input signal. The binary-scaled array of unit capacitors C_U is pre-charged to the level of supply voltage. The charge in this capacitor array will be used to provide the feedback DAC function in the SAR ADC, functions as the “reference voltage” in a charge-redistribution SAR. Since this pre-charging happens before the actual A/D conversion, it does not depend on the input signal and thus no tough constraints are imposed in this “reference” [1].

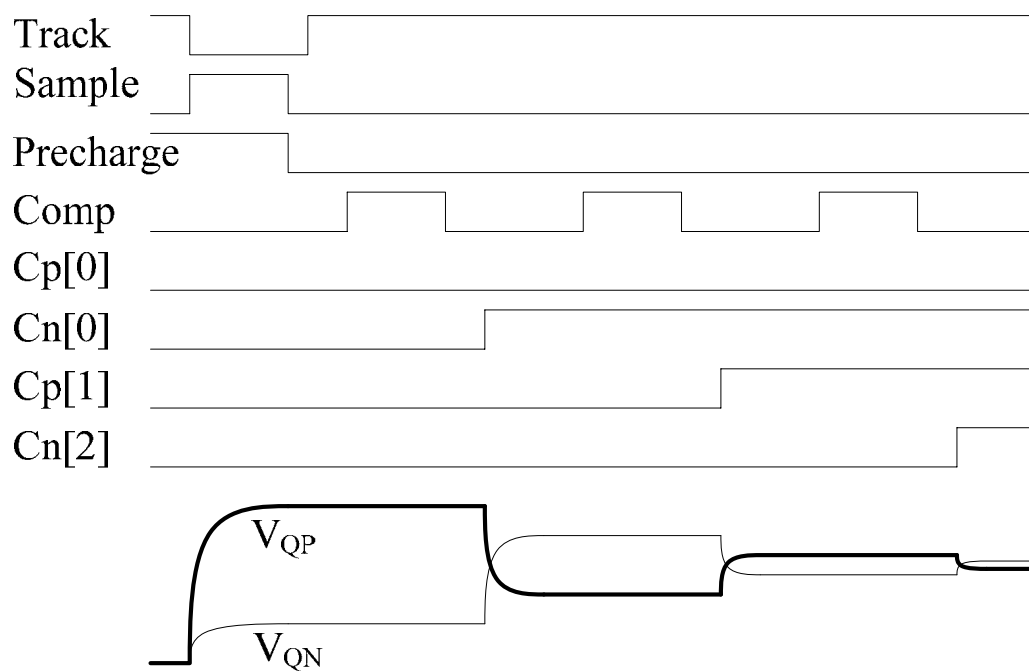


Figure 4-2 Charge-sharing SAR ADC waveforms [1]

The conversion process starts with the sampling of the input signal. Therefore, S_T opens to hold the input and when S_S closes, half of the charge in C_T is transferred to C_S in a sample PCS action. The transfer time is proportional to the on-resistance of the switch and therefore, is very short. After sampling, S_S opens and S_T closes, so the

input signal can be tracked again. The sampling process typically $< 2\text{ns}$, so abundant time is available for tracking and setting problems occur [1].

To determine the MSB, the comparator is activated, and after its decision a charge equal to half of the input range is added to or subtracted from the sampled charge. Therefore, one of the signals $C_p[0]$ or $C_n[0]$ goes high, and the other stays low. This is again a passive process that does not take any power and yet settles very fast in an advanced technology. The total charge stored now is equal to $C_s \cdot V_{in} / 2 \pm C_{MSB} \cdot V_{DD}$ and is distributed proportionally across the 2 capacitors. The next bits are determined in a similar operation: the comparator determines the sign of the charge for each iteration, and the binary-scaled pre-charged capacitors are connected one by one to make the total charge to zero [1].

4.2.1 Capacitor Array

Two constraints determine the size of the capacitors. For device mismatch, the total size of the capacitor array must be at least 2pF . Much more stringent however, is the size of the unit capacitor, which must be 2^{N-1} times smaller than the largest value and thus turns out to be only 8fF . As this is not practical, the capacitor array, shown in Figure 4-3 is designed. A unit capacitor of 64fF is used, but for the 3LSBs, which one is pre-charged to only a fraction of the fully supply voltage. Again, passive charge-sharing is used, as shown by the waveforms in Figure 4-3 [1].

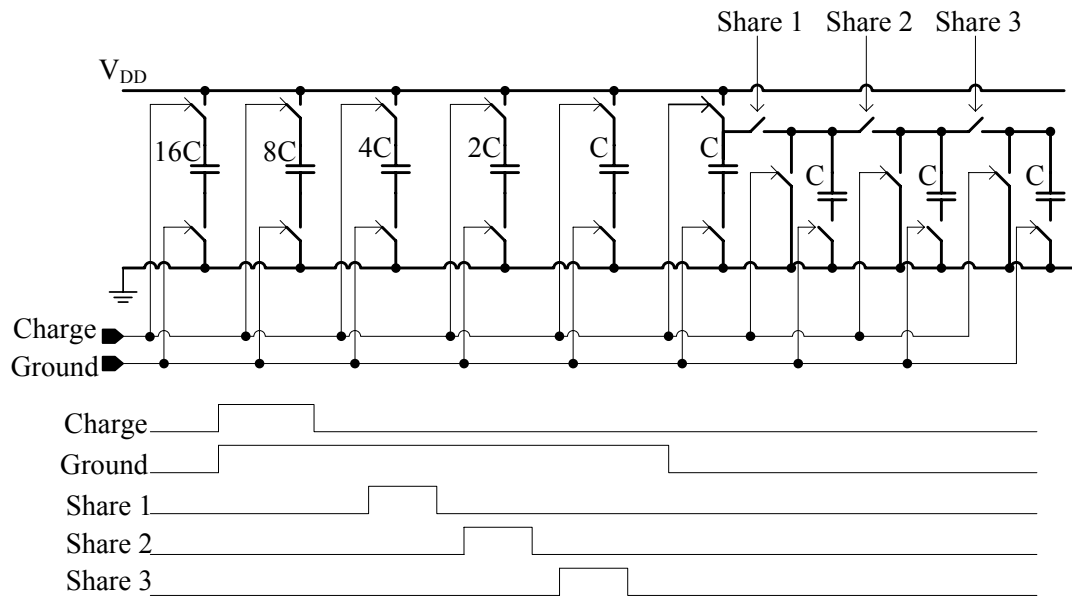


Figure 4-3 Capacitor array and control waveform [1]

4.2.2 Comparator

The comparator is shown in Figure 4-4. It is designed based on [4] and has a programmable capacitor array to calibrate its offset voltage. The comparator does not consume any power when inactive, and thus the power consumption of the ADC scales linearly with the sampling frequency. The output Valid indicates when the comparison is done and it is used by the asynchronous SAR controller as a trigger to continue the conversion process by closing one of the sharing switches [1].

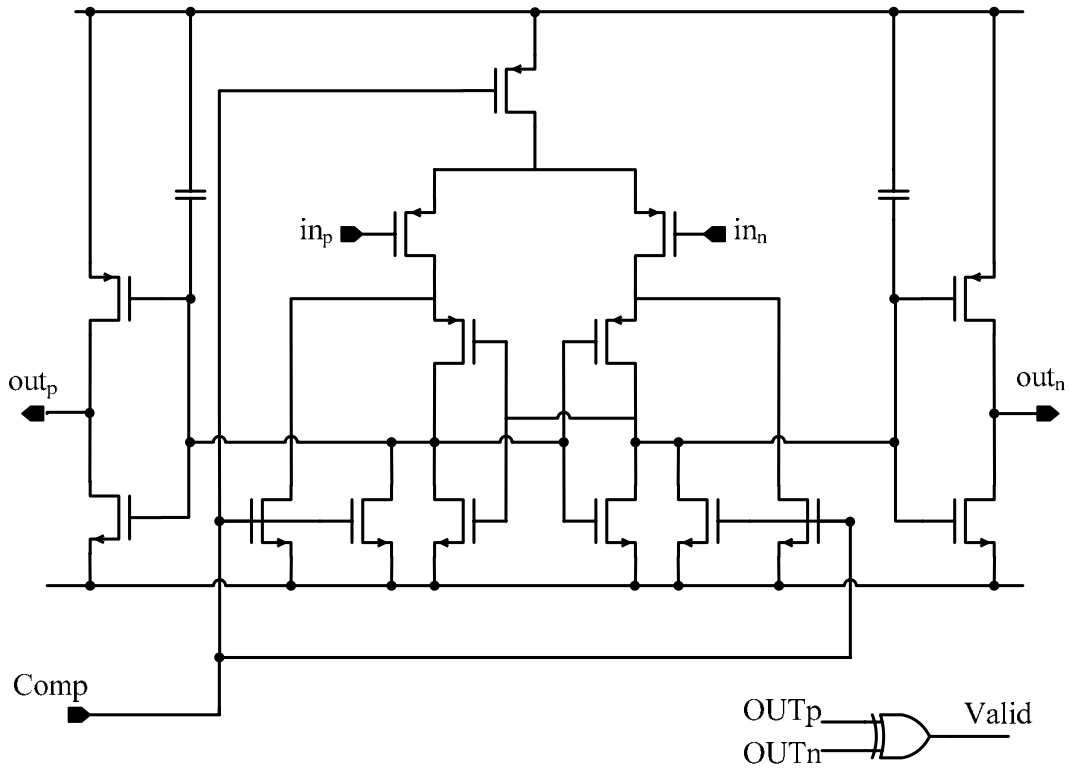


Figure 4-4 Comparator circuit schematic [1].

4.2.3 Digital Control Circuit

To avoid the need for a high-speed clock (and its associated power consumption), an asynchronous controller is implemented. On the rising edge of the input clock, a pulse in the tracking/sampling signal is generated as shown in Figure 4-2. The width of this pulse ($\sim 2\text{ns}$) is proportional to the RC time constant of a MOS switch and a MIM capacitor to track the time required for the charge-sharing sampling process over process variations. Then the conversion process starts that repetitively: activates the comparator, waits for the Valid signal, closes one of the sharing switches, and waits for another delay ($\sim 1\text{ns}$) while the charge-sharing settles. When this has happened 9 times, the output bits are toggled [1].

4.3 The Architectures of Proposed Passive Charge-Sharing SAR ADC

A large capacitance, 10 pF, is used at the front end of the PCS SAR ADC to track the input signal. This capacitance will cause a large capacitive load for the front end circuit and difficult to integrate with other system on a chip. Most of the charge-redistribution SAR ADCs also have this problem.

A suitable capacitance, 1.25 pF, is used in the proposed PCS SAR ADC. In the Figure 4-5, the full architecture of the proposed PCS SAR ADC is shown, and it is the same as the architecture shown in the section 4.2, besides the capacitor array.

The proposed capacitor array is shown in Figure 4-6; two capacitors are used in each stage with the same capacitance. One node of each capacitor is connected to the common voltage V_{CM} to avoid the voltage of the capacitor to float. All the capacitors in the capacitor array have the same size, instead of binary-scaled. For device matching consideration, the total capacitance of the capacitor arrays must be at least 2pF, hence the unit capacitance is 250fF. For this reason, the capacitor size of each tracking and sampling capacitor is just 1.25pF. Therefore, the core area of this ADC will be very small and be suitable to integrate with other circuits.

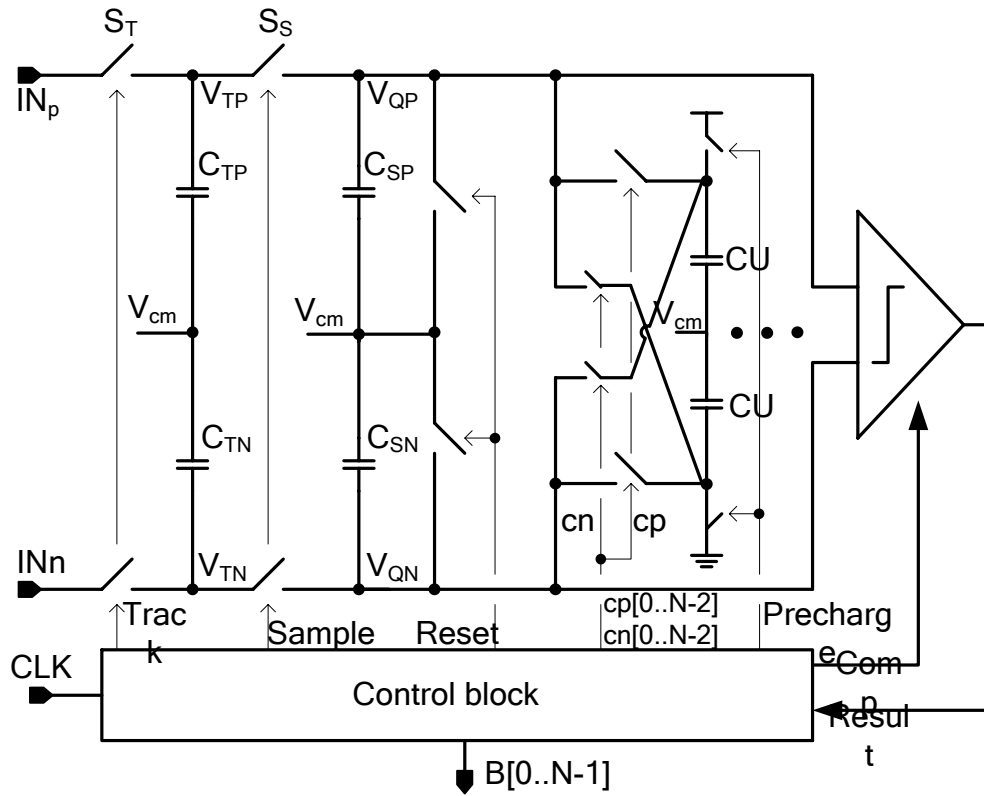


Figure 4-5 Proposed passive charge-sharing SAR ADC

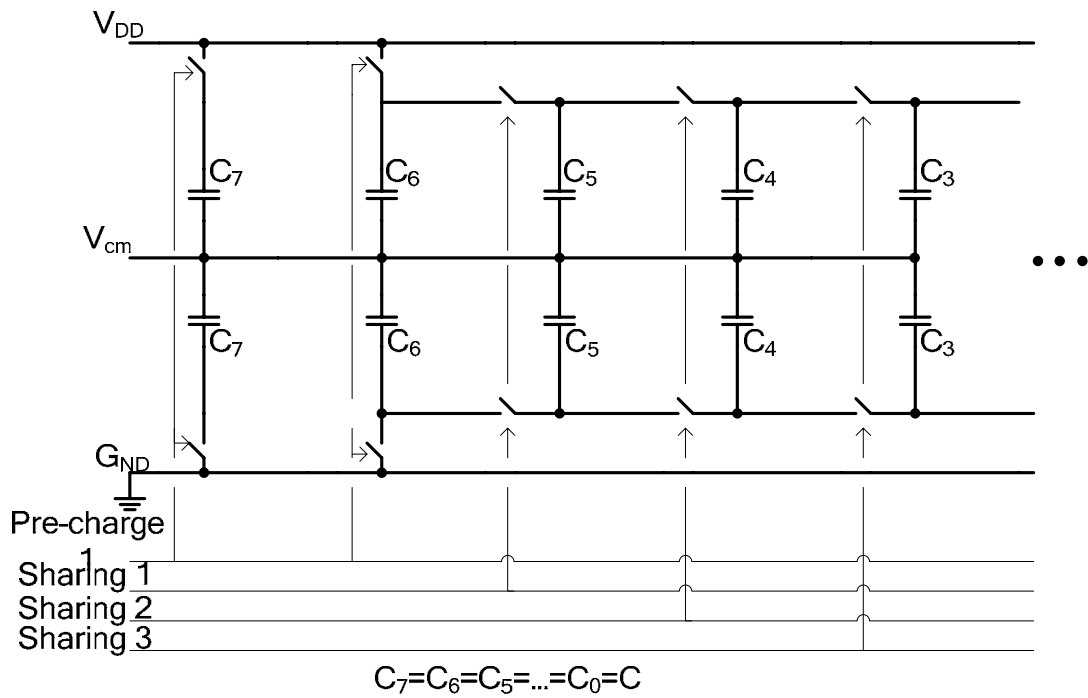


Figure 4-6 Capacitor arrays of proposed passive charge-sharing SAR ADC

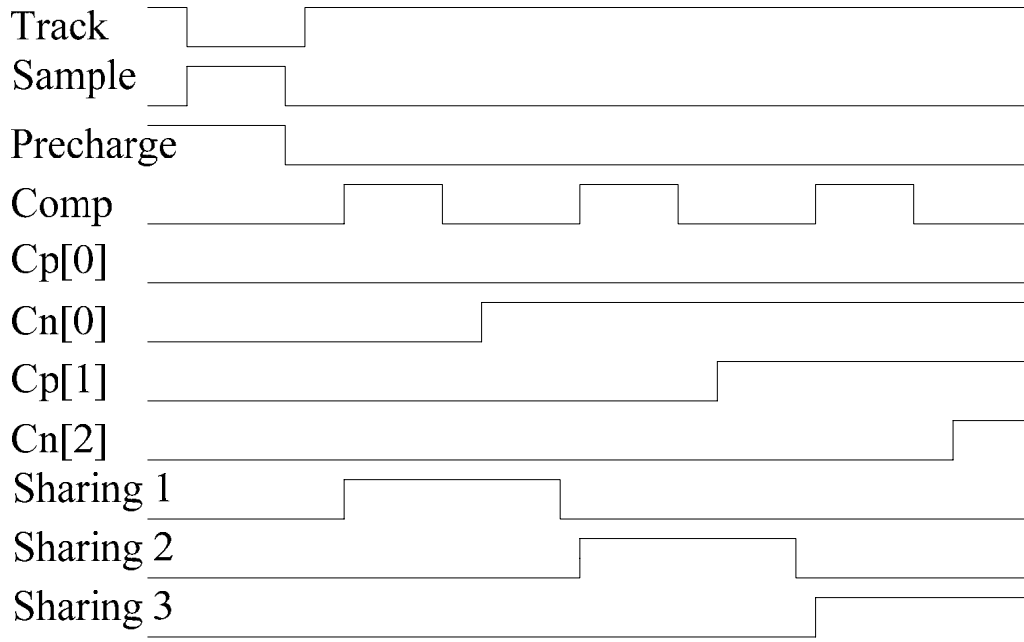


Figure 4-7 Proposed charge-sharing SAR ADC control waveforms

4.3.1 Capacitor Array

In section 4.2, the charges stored in capacitor array is binary-scaled, $16C \cdot V_{DD}$, $8C \cdot V_{DD}$, ..., $1/4C \cdot V_{DD}$. In the proposed charge-sharing capacitor arrays, only the first and second stages are pre-charged to the supply voltage, while the charges in the C_7 and C_6 have the same value, $C \cdot V_{DD}$. After the pre-charging phase, the charge-sharing between the tracking capacitor C_T and sampling capacitor C_S have been completed. Then the comparator will compare the voltage of V_{QP} and V_{QN} to determine the MSB. At the same time, the switch, Sharing 1, will be closed and half of the charges stored in the capacitor C_6 will share to the capacitor C_5 . consequently, the charges stored in the capacitor C_6 will become $1/2C \cdot V_{DD}$, half of the capacitor C_7 . The control waveforms are shown in the Figure 4-7.

For the same movement, the total charges stored in each capacitor will become binary-scaled , $C_7 = C \cdot V_{DD}$, $C_6 = 1/2 C \cdot V_{DD}$, $C_5 = 1/4 C \cdot V_{DD}$, ..., $C_1 = C_0 = 1/64 C \cdot V_{DD}$, finally.

In the pre-charging phase, only the capacitor C1 and C2 are charged to the level of supply voltage, so the total power consumption of the proposed capacitor arrays is roughly quarter of the conventional.

4.3.2 Comparator

The capacitance values at the comparator input node are very large generally, so the kickback noise can be neglected. However, in this design the capacitances are decreased, only 2~3pF, but the kickback noise will become very serious. In Figure 4-9, the voltages of the sampling capacitor, V_{QP} and V_{QN} , are shown during the successive approximation process, where a dynamic latched comparator is used. In the right of figure 4-9, some glitches caused by kickback noise are shown during the conversion of 5 LSBs, which is enlarged in Figure 4-10.

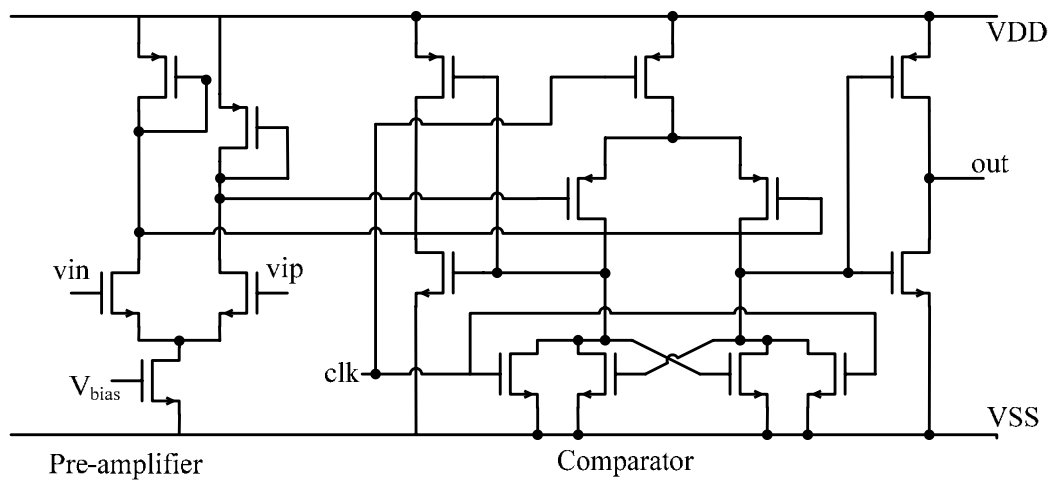


Figure 4-8 Proposed comparator with pre-amplifier

Therefore, it needs a pre-amplifier to keep the capacitor from the kickback noise. A proposed comparator with a pre-amplifier is presented in the Figure 4-8. With the pre-amplifier, the conversion of 5 LSBs is shown in the Figure 4-11, where the glitches caused by kickback noise disappear.

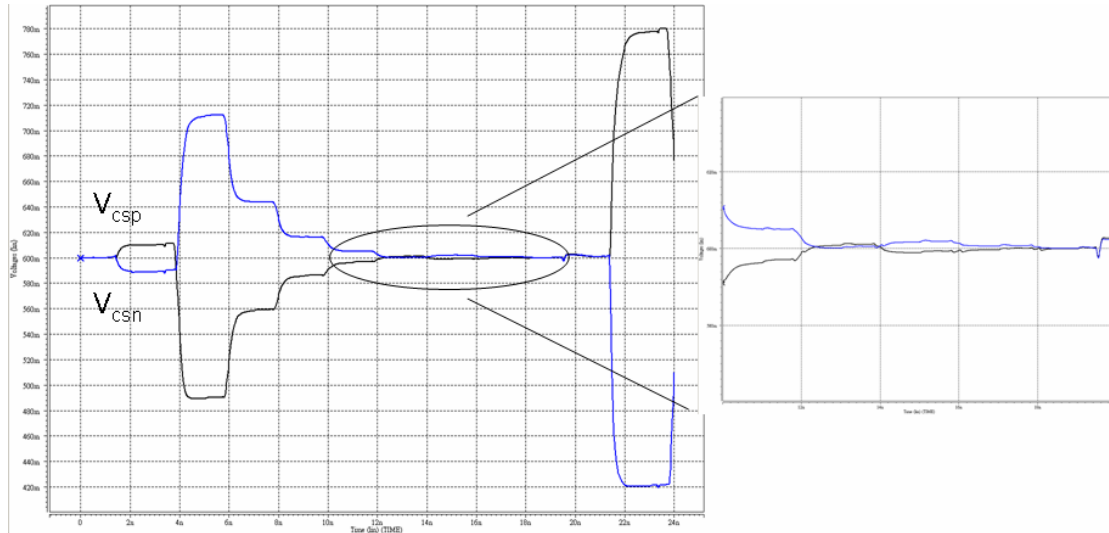


Figure 4-9 Work function at the comparator input node

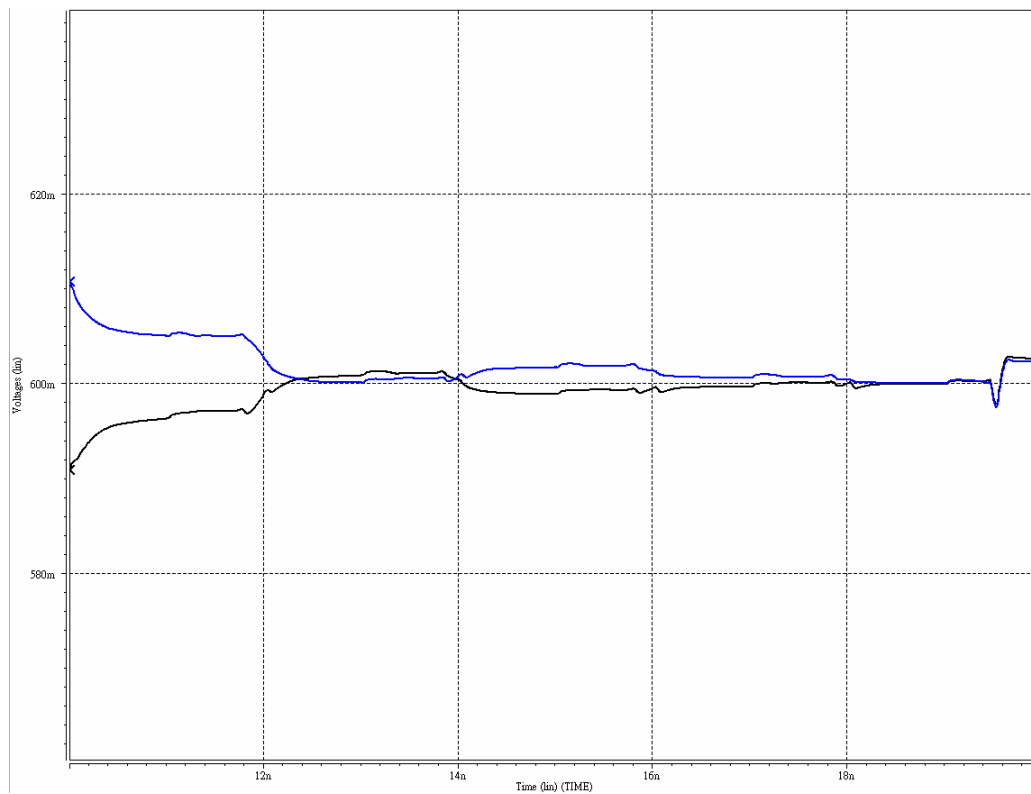


Figure 4-10 Conversion of the 5 LSBs without pre-amplifier

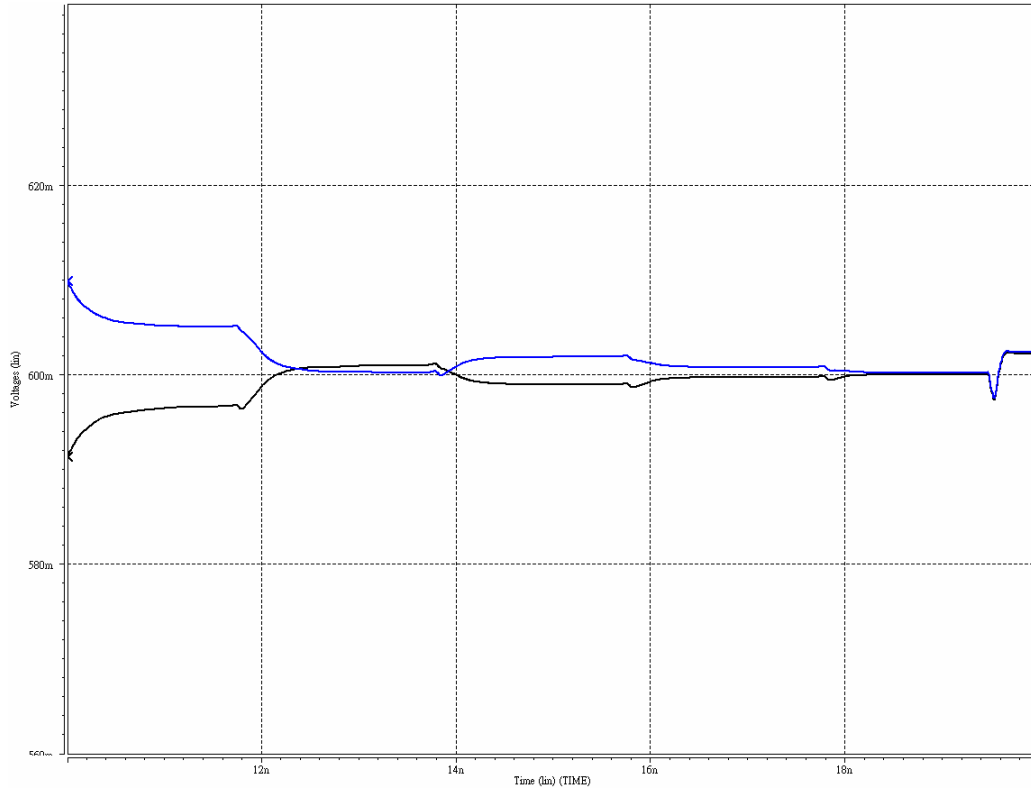


Figure 4-11 The conversion of the 5 LSBs with pre-amplifier

4.3.3 Digital Control Logic

In the conventional PCS SAR ADC, an asynchronous digital controller was used to increase the speed. However, this design uses the same size capacitor in the proposed capacitor array, so the settling time of each stage is equal. Consequently, a synchronous digital controller is applied in the proposed ADC and it can achieve 50MS/s conversion rate as the conventional.

4.4 Simulation Results

4.4.1 Functional Simulation

During the successive approximation process, the fully differential voltage V_{QP} and V_{QN} will be added or subtracted until the difference voltage between V_{QP} and V_{QN} is less than $1/2$ LSB. Figure 4-12 presents the conversion process of code 255, the V_{QP} (V_{QN}) is always subtracted (added) during the successive approximation process. After the conversion process, the V_{QP} and V_{QN} need to be reset to the common voltage V_{CM} .

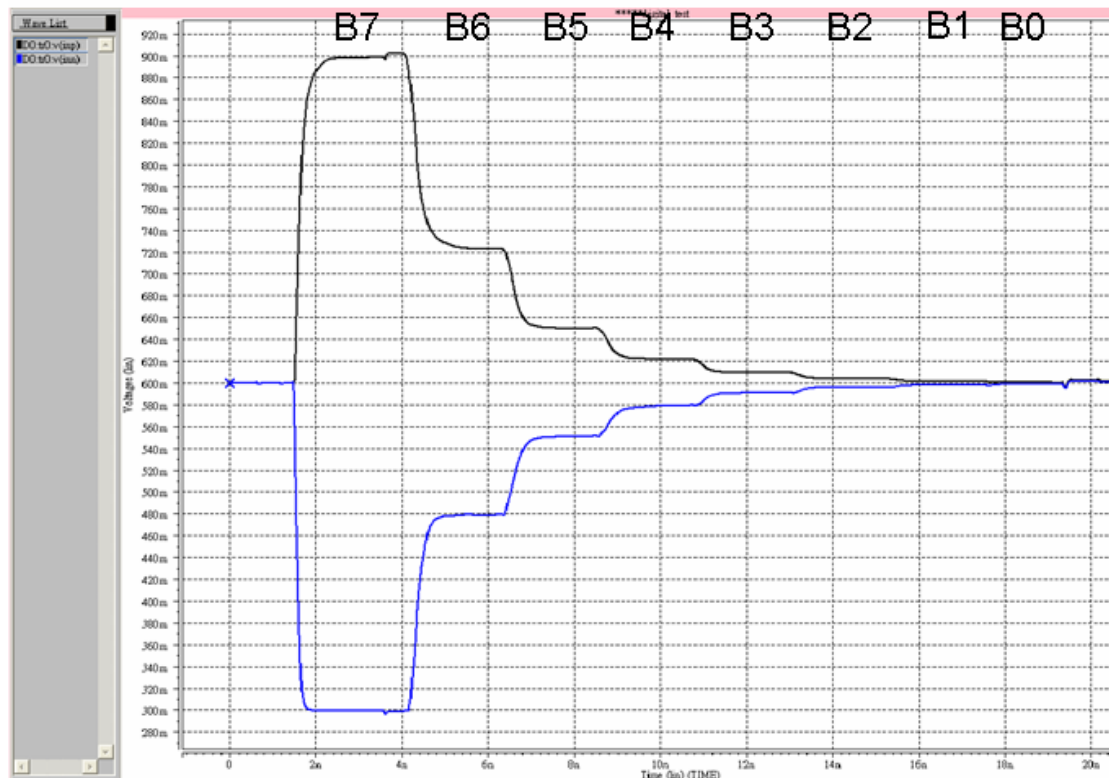


Figure 4-12 Conversion waveform of proposed ADC

4.4.2 Pre-Layout Simulation

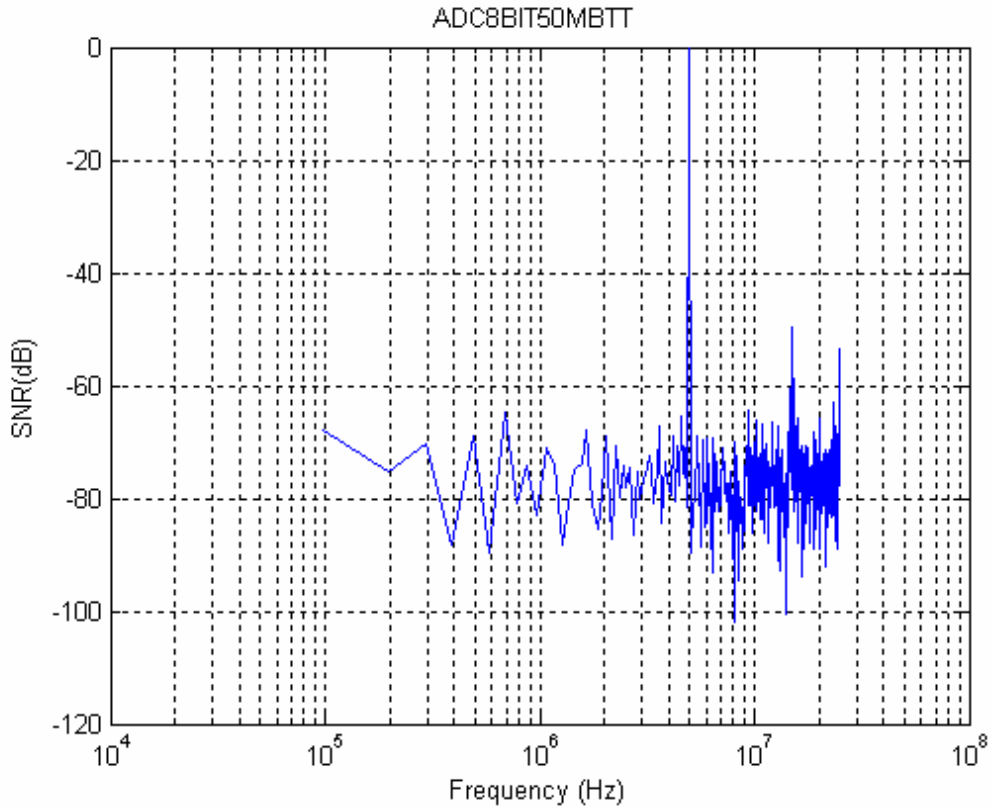


Figure 4-13 FFT simulation result of the output code with $f_{in} = 5\text{MHz}$ and 50MS/s conversion rate (SNDR = 45.0dB and ENOB = 7.18 bit)

This 8-bit 50MS/s PCS SAR ADC uses TSMC $0.13\mu\text{m}$ 1P8M technology. For FFT analysis, 512 points are sampled with 5MHz input and 50MS/s conversion rate, as shown in Figure 4-13, where the SNDR = 45.0 dB and ENOB = 7.18bit.

Table 4-1 lists the simulation results of the proposed ADC in different process corners, the item FS means fast NMOS transistors and slow PMOS transistors, and the other items can be explained in the same way.

The charge injection of the switch affects the signal to noise and distortion ratio critically. It relies on some further techniques to cope with this problem.

Input / Corner	SNDR / ENOB
5MHz / TT	45.0 / 7.18
5MHz / FF	44.71 / 7.13
5MHz / SS	43.68 / 6.96
5MHz / FS	43.94 / 7.01
5MHz / SF	44.02 / 7.02

Table 4-1 Corner deviation simulation

The total power consumption of the proposed PCS SAR ADC is 285 μ W, it composed of each component as follows: DAC 27%, comparator 28%, and digital 45%, as shown in Table 4-2. Most of the power is consumed by the digital control circuit, because a lot of switches are used in the capacitor arrays, which need complicated controller to control. However, lots of the power consumption can be saved by using the proposed capacitor array, so the power consumed in the capacitor arrays is only 80 μ W.

Component	Power(μ W)
Comparator	83
Capacitor array	80
SAR	131
Total (Avg.)	294

Table 4-2 Power consumption of each component

Specifications	Results
Supply Voltage	1.2V
Sampling Frequency	50 MHz
Input Common-Mode Voltage	0.6V
Input Range	0.96 V _{p-p}
Output Form	Parallel out
ENOB	6.9~ 7.2
Power	294 μ W (Avg.)
Resolution	8 bits
Process	TSMC0.13 μ m 1P8M Process

Table 4-3 The specification of the 8-bit 50MS/s PCS SAR ADC

After all the pre-layout simulations are introduced, specifications of the proposed SAR ADC are summarized in Table 4-3. The signal to noise ratio with 5MHz input is 45.0dB at 50MS/s.

Chapter 5

Conclusion and Future Works

5.1 Conclusion

Two successive-approximation analog-to-digital converters are presented in this thesis. First, an 8-bit 27MS/s SAR ADC, which is designed for DTV applications, is proposed in the chapter 3. The first challenge of this design is to increase the conversion rate of the SAR ADC to meet the requirement of DVT applications. To cope with the limited bandwidth of the switched-capacitor array, a novel RC time constant array is used to raise the conversion rate up to 27 MS/s. A self timing controller mechanism [14] is utilized in the proposed comparator to save approximately 50% power consumption. Therefore, the total power consumption is 385 μ W and the averaged energy consumption per conversion step is just 105 fJ.

A new passive charge-sharing (PCS) SAR ADC is shown in the chapter 4. The proposed PCS SAR ADC, which has the same size capacitors in the capacitor array, can reduce the size of the tracking and sampling capacitors to 1~2pF, and consequently reduce the power consumption. With pre-charge sharing mechanism the proposed PCS SAR ADC does not require redundant pre-charging time, so the conversion rate can achieve 50MS/s without any asynchronous time controller. The total power consumption is 294 μ W and the FOM is 41 fJ/conversion-step.

5.2 Future Works

The proposed PCS SAR ADC has serious charge injection noise. It relies on some further techniques to cope with this problem.

The architecture of the successive approximation ADCs is very simple, which is just composed of an active component, the comparator, so the power consumption of SAR ADCs is very economical. However, for an N-bit SAR ADC $N+K$ clock cycles are required to complete a conversion, it is difficult to attain a high conversion rate. Therefore, how to increase the conversion rate for wide applications is still a challenging research task.

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